Multivariate Ranks Using Optimal Transport

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Submitted for the Master of Science, London School of Economics,
University of London

Acknowledgement

Fourteen months ago, I was just a freshly graduated undergraduate student; fourteen months later, I am already about to get my MSc degree, and in just one month I will be starting my PhD journey. It all feels a bit surreal: everything that has happened in these fourteen months still seems so vivid, as if it all just took place yesterday.

The first year has truly been fulfilling. Most of my time was spent on PhD applications, coursework, projects, exam preparation, and writing my MSc dissertation. In terms of effort, I would say this year comes second only to my freshman year in college (if we start counting from after the college entrance exam).

First and foremost, I want to thank my parents. They have always offered me unconditional material and emotional support in all my decisions. Thanks to them, I, someone born into a common but loving and supportive family, have had the opportunity for this first year abroad and my very first study experience overseas. I am also grateful to my other family members for their care and support.

I am deeply thankful to my supervisor, Prof. Wang, who has always given me the greatest support, encouragement, and guidance throughout the entire PhD application and MSc dissertation writing process. During my most anxious times, he managed to ease my worries in the most effective way. Despite his remarkable ability, he never showed any impatience even when I asked very basic or seemingly silly questions. Whether as a supervisor, a mentor or a teacher, everyone I know speaks highly of him (even including random posts I came across on a social media), and I feel truly fortunate to be his PhD student.

I would also like to thank all the teachers and classmates who helped me during my applications: my referees Dr. Shanshan, Dr. Yanfie, and Prof. Huiwen, who have consistently supported me with recommendation letters throughout the whole process of applying for both MSc and PhD programs; senior students like Dr. Xuzhi, Di, and Kaixin, who patiently answered many of my questions.

Lastly, I am grateful for all the friends I met in the last few months: my six flatmates-Carolina, Haomin, Ines, Ningning, Siyi and Sumi, it was my privilege to live with you for ten months, and I will never forget the moments when we shared meals, sang, drank, went clubbing, played games, and talked about everything. To my group project members, working side by side with you on each project has been a wonderful memory (even though the process was sometimes a bit painful). And to the other friends from all over the world, it has been my honor to meet you at LSE.

Summary

Nonparametric statistical tests are widely used due to their minimal assumptions on data distributions, and rank-based methods are especially valued for robustness against outliers and heteroscedasticity. However, generalizing ranks from univariate to multivariate data is challenging because there is no natural ordering in higher dimensions. Existing multivariate rank definitions like component-wise, spatial, and depth-based ranks have limitations such as distribution dependence and non-uniqueness, which restrict their effectiveness in distribution-free testing.

To address these issues, this dissertation introduces a novel multivariate rank framework based on optimal transport theory. By mapping data to a known reference distribution through the Monge-Kantorovich transport map, this approach yields distribution-free ranks with strong theoretical guarantees. Two practical implementations, cell decomposition and Halton sequences with the Hungarian algorithm, are proposed, with the latter offering improved efficiency and scalability.

Building on this foundation, we develop three new nonparametric tests: the Rank Minimum Spanning Tree test, the Rank Anderson Darling test, and a multivariate Wald Wolfowitz runs test. These tests operate in rank space, allowing effective detection of distribution differences without strong parametric assumptions.

Simulation studies demonstrate that the proposed methods are robust and accurate across various scenarios. The Rank MST test, in particular, outperforms classical methods such as the Energy Distance test in capturing complex distributional features. Real data applications confirm the practical utility of our Rank MST test in distinguishing groups with high significance.

Overall, our work presents a feasible and effective framework for multivariate ranks and related nonparametric testing.

Codes that are used in this dissertation can be viewed here: CODES-LINK.

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1 Introduction

1.1 Ranks in Nonparametric Statistical Testing

Compared to parametric testing, nonparametric testing enjoys the convenience of minimal reliance on assumptions about probability distributions. Within the category of nonparametric testing statistics, rank statistics play a vital role among them. Ranks in $\mathbb{R}$ rely on the ordering of real numbers, which in other words, reveals the nature of rank statistics is built on the relative magnitudes of data rather than their absolute values.

With this feature, rank statistics possess advantages in various cases when carrying out statistical testing. When dealing with data containing outliers, rank statistics are not influenced by some certain outliers, while several classical parametric test statistics are severely affected by outliers. A great amount of parametric test statistics require homoscedasticity of the data, for instance, Student's t-test and ANOVA, etc. However, cases are that homoscedasticity is always violated in real data analysis. Rank statistics, instead, are robust to heteroscedasticity. Moreover, in the context of small-sample analysis, some classical parametric test statistics may not be effective due to the instability of sample moments, while rank statistics showcase effectiveness in that case.

1.2 Multivariate Ranks

Moving to performing tests on multivariate data, the first problem that arises is how to define multivariate ranks. Unlike univariate ranks that can be accessed by simply ordering on the real line, multivariate ranks do not enjoy the same property, since there is no natural ordering in $\mathbb{R}^d$ for $d > 1$.

To tackle this issue, different notions of multivariate ranks have been proposed in the statistical literature. These include component-wise ranks, spatial ranks, depth-based ranks. Component-wise ranks involve ranking the univariate marginals separately, spatial ranks determine the relative positioning of data points based on spatial distance from the distribution center, and depth-based ranks utilize statistical depth concepts. We will talk about these notions in the following section.

A key drawback of the above multivariate rank concepts is their lack of distribution-free properties, making them less attractive compared to their univariate counterparts. This brings up the critical question of how to define multivariate ranks that support distribution-free testing methods.

Recently, the notion of multivariate ranks defined using optimal transport was proposed. Essentially, this notion leverage the geometry of the optimal transport problem

with the squared Euclidean metric as the cost function, where under some mild conditions the optimal maps are gradients of convex functions. Specifically, the optimal transport-based multivariate ranks can be viewed as an assignment problem, and thus can be solved by some well-studied algorithms. With this definition, we are able to generalize various statistical tests into their multivariate ranks version.

2 Related Work

In this chapter, We begin with revisiting several previously proposed definitions of multivariate ranks, followed by an introduction and the motivation for the optimal transport-based multivariate ranks, before a brief overview of the optimal transport concept.

Prior to that, we first introduce the concept of distribution-free, which is the key concept discussed throughout this dissertation. We adopt the intuition given in Hallin et al. (2021a). Suppose we are analyzing a random sample, the ranks are called distribution-free if they contain no information about the distribution that generates the sample. This is equivalent to say that if we start with a different distribution that generates the sample, the distribution of the ranks will remain the same.

2.1 Definitions of Multivariate Ranks

We start with a literature review of various definitions of multivariate ranks, including component-wise ranks, spatial ranks and depth-based ranks.

2.1.1 Component-wise Ranks

The notion of component-wise ranks goes back as far as Bickel (1965) and Puri and Sen (1966), after which Puri and Sen (1971) integrated a series of component-wise ranks based statistics and proposed inference procedures under them. The basic idea of component-wise ranks is, instead of dealing with the tricky problem of ordering data in $\mathbb{R}^d$, it cares about each component. Given a random sample $X_1, \ldots, X_n$ in $\mathbb{R}^d$, where each X_i is denoted as $X_i = (X_{i1}, \ldots, X_{ij}, \ldots, X_{id})$, the component-wise rank of X_i is defined as

$$R(X_i) = (r_{i1}, \ldots, r_{ij}, \ldots, r_{id})^T$$

where r_{ij} is the univariate rank of X_{ij} among $\{X_{1j}, \ldots, X_{nj}\}$. In fact, component-wise ranks are closely connected to copula transformation. The definition above can also be viewed as

$$R(X_i) = (F_1(x_{i1}), \ldots, F_j(x_{ij}), \ldots, F_d(x_{id}))^T$$

where F_j is the jth marginal empirical CDF of $X = (X_1, \ldots, X_j, \ldots, X_d)$.

Under this definition, the component-wise ranks relies on specific choices of coordinates and thus, are not invariant under simple transformations of the data. In fact, as discussed in Ilmonen and Paindaveine (2011) and Hallin and Mehta (2015), the distri-

bution of the component-wise ranks depends highly on marginal distributions: unless the underlying distribution has independent components in the chosen coordinates, component-wise ranks is not distribution-free.

2.1.2 Spatial Ranks

This class of ranks initially appeared in Möttönen and Oja (1995) and Möttönen et al. (1997). Inspired by the fact that in univariate case, ranks indeed measure each observation's relative position within the sample, it generalized this idea into multivariate case by calculating each observation's position to the center of the sample, with the center usually calculated by spatial median

$$R(X_i) = ||X_i - \widehat{m}||$$

where $\widehat{m} = \text{argmin}_{m \in \mathbb{R}^d} \sum_{i=1}^{n} ||X_i - m||$ is the spatial median of the sample, and $|| \cdot ||$ is the $L1$ norm in $\mathbb{R}^d$.

Similarly, Oja and Randles (2004) and Oja (2010) defined their spatial ranks by

$$R(X_i) = \frac{1}{n-1} \sum_{j \neq i} U(X_i - X_j)$$

where

$$U(X) = \begin{cases} \frac{X}{||X||} & X \neq 0 \\ 0 & X = 0 \end{cases}$$

Instead of measuring each data point's position to the center of the sample, they measure each data point's position to every other data points using the notion of sign. The same idea was also developed in high dimension, for example, by Chakraborty and Chaudhuri (2014). Generally speaking, this genre of multivariate ranks still does not hold the property of distribution-free.

2.1.3 Depth-based Ranks

Basically speaking, the main idea of depth-based ranks is, for a probability measure P and a random variable X in $\mathbb{R}^d$, we define a function $D(x; P)$ to provide a P-based center-outward ordering of any point x in $\mathbb{R}^d$. The center-outward property reveals that the basic thought in depth-based ranks lies in finding a center, which represents the deepest, of the distribution (or empirically speaking, finding a center of the data cloud). It calculates the depth of each point to the center: the larger the depth value, the closer it is to the center; while the smaller the depth value, the farther it is from

the center. Then it establishes a center-to-outward ordering of the data using depth contours. We recommend Zuo and Serfling (2000) for a rough understanding of the concept and some previously proposed depth concepts.

The most original and classical statistical depth concept was proposed by Tukey (1975), which is called Tukey depth (or halfspace depth). The population version of the Tukey depth is defined as

$$\begin{aligned} D(x;P) &= \inf_{v\in\mathbb{R}^d,||v||=1} P(v^T(X-x)\geq 0) \\ &= \inf\{P(H) : H \text{ is a closed halfspace that contains } x\} \end{aligned}$$

It is the probability mass of the smallest halfspace in $\mathbb{R}^d$ that contains x. Dutta et al. (2011) showed that properties of the Tukey depth include: Tukey depth takes value in $\mathbb{R}$ and specifically, in $[0, 1/2]$; the contours of the Tukey depth are continuous and convex; level sets (point sets with each point's Tukey depth equals or greater than a given $d \in \mathbb{R}$) of the Tukey depth are closed, convex and nested outwards as d decreases. Chernozhukov et al. (2017) showed that due to the convexity of it contours, the Tukey depth cannot reflect non-convex features in the geometry of the underlying distribution. We take a banana-shaped distribution as an example. From the definition of the Tukey depth, we can easily tell that $D(A)$ is greater than 0 while $D(B)$ is 0. Both points are on the margin of the data cloud but their Tukey depths are far different. In that case, the Tukey depth loses a great deal of information in our distribution.

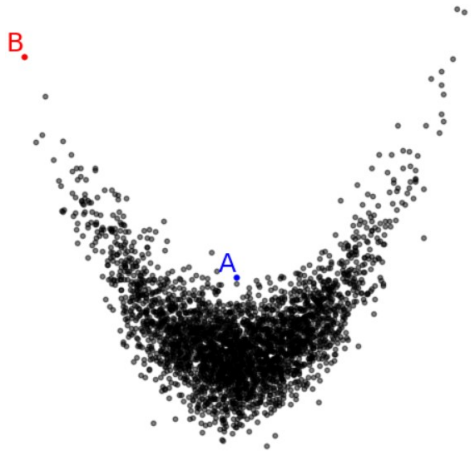


Figure 1: A Banana-shaped Distribution Example

A banana-shaped distribution drawn from $X \sim N(0,1)$ *and* $Y \sim X^2 + \epsilon$ *where* $\epsilon \sim N(0,1)$.

Ever since then, a great amount of depth-based ranks were proposed. Simplicial depth was brought up by Liu (1990), and the idea of it depends on simplexes in $\mathbb{R}^d$. Suppose we have a random sample $X_1, \ldots, X_{d+1}$, and denote the simplex with vertices $X_1, \ldots, X_{d+1}$ by $S[X_1, \ldots, X_{d+1}]$, which refers to the set of all points in $\mathbb{R}^d$ that are

convex combinations of $X_1, \ldots, X_{d+1}$. Then the simplicial depth of a point $x \in \mathbb{R}^d$ is defined as

$$D(x; P) = P(x \in S[X_1, \ldots, X_{d+1}])$$

Mahalanobis depth and majority depth were considered in Liu and Singh (1993). Mahalanobis depth is defined using Mahalanobis distance. For a point $x \in \mathbb{R}^d$ that is drawn from $X \sim P$, suppose $\mathbb{E}(X) = \mu$ and Σ is the variance-covariance matrix of X, the Mahalanobis distance of x is defined as

$$d_M(x) = (x - \mu)\Sigma^{-1}(x - \mu)$$

And thus, the Mahalanobis distance of x is defined as

$$D(x; P) = \frac{1}{1 + d_M(x)}$$

As for the majority depth, suppose we have a random sample $X_1, \ldots, X_p$, it is defined as

$$D(x; P) = \min_{H \in \mathcal{H}_x} P(X \in H) = \min_{H \in \mathcal{H}_x} \frac{1}{p} \sum_{i=1}^{p} \mathbf{1}(X_i \in H)$$

where $\mathcal{H}_x$ is the set of all hyperspaces that are bounded by x. It can be seen that the majority depth is pretty similar to the Tukey depth, while the Tukey depth takes values from $[0, 1/2]$, the majority depth takes values from $[1/2, 1]$.

Other depth concepts also includes the projection depth proposed by Donoho and Gasko (1992), etc. Zuo and He (2006) investigated depth-based rank sum statistics and related tests, López-Pintado and Romo (2009) generalized the idea into functional data, and Zuo (2021) conducted a study on the statistical depth in a regression context. Also see Mozharovskyi et al. (2020), Singh et al. (2022), and Mosler and Mozharovskyi (2022) for some state-of-the-art studies.

Generally speaking, depth-based ranks still do not enjoy the property of distribution-free. Moreover, just as mentioned earlier, they lack some appealing properties that univariate ranks holds. For example, the non-uniqueness of ranks (the feature that different data points may have a same rank) in high dimension leads to the difficulty in ordinal comparison and the inconsistency in empirical ranks when different algorithms are applied to the same dataset. In addition, depth-based ranks can be computationally expensive. Chan (2004) showed that the computational complexity of the Tukey depth is $O(n^{d-1})$. The computational complexity of both Mahalanobis depth and majority depth exhibits exponential growth as the dimensionality increases.

2.2 Optimal Transport-based Multivariate Ranks

An important breakthrough was made by the pioneering work Chernozhukov et al. (2017) and Hallin et al. (2021a). Derived from the concept of depth-based ranks that we discussed in the last section, they introduced the Monge-Kantorovich depth, which essentially is transport maps between distributions, to provide a novel concept of multivariate ranks, quantiles and signs. They discussed this topic from the prospective of statistical inference and came up with a new definition of multivariate distribution and quantile functions. Moreover, Ghosal and Sen (2022) moved one step forward by considering the topic in nonparametric testing and illustrated how to construct specific nonparametric tests using optimal transport-based ranks.

To motivate optimal transport-based multivariate ranks, we start with univariate ranks. Ranks in one dimension can be viewed essentially as an assignment problem. Given a random sample $X_1, \ldots, X_n$ on $\mathbb{R}$, the rank map assigns the observations to elements in the set $\{1/n, 2/n, \ldots, n/n\}$, which can be interpreted as solving the optimization problem

$$\begin{aligned}\widehat{R} &= \text{argmin}_{\sigma=(\sigma(1),\ldots,\sigma(n))\in S_n} \sum_{i=1}^{n} \left| X_i - \frac{\sigma(i)}{n} \right|^2 \\ &= \text{argmax}_{\sigma=(\sigma(1),\ldots,\sigma(n))\in S_n} \sum_{i=1}^{n} \sigma(i) X_i\end{aligned}$$

where S_n is the set of all permutation of $\{1, 2, \ldots, n\}$ (see Villani (2021)). Applying the rearrangement inequality that for every choice of real number $x_1 \leq x_2 \leq \cdots \leq x_n$ and $y_1 \leq y_2 \leq \cdots \leq y_n$, and for every permutation σ of $\{1, 2, \ldots, n\}$, we have

$$x_1 y_{\sigma(1)} + x_2 y_{\sigma(2)} + \cdots + x_n y_{\sigma(n)} \leq x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

Thus $\frac{\widehat{R}(i)}{n} = \widehat{\sigma}(i)$ is actually the rank of X_i in its random sample. This idea is illustrated below in Figure 2.

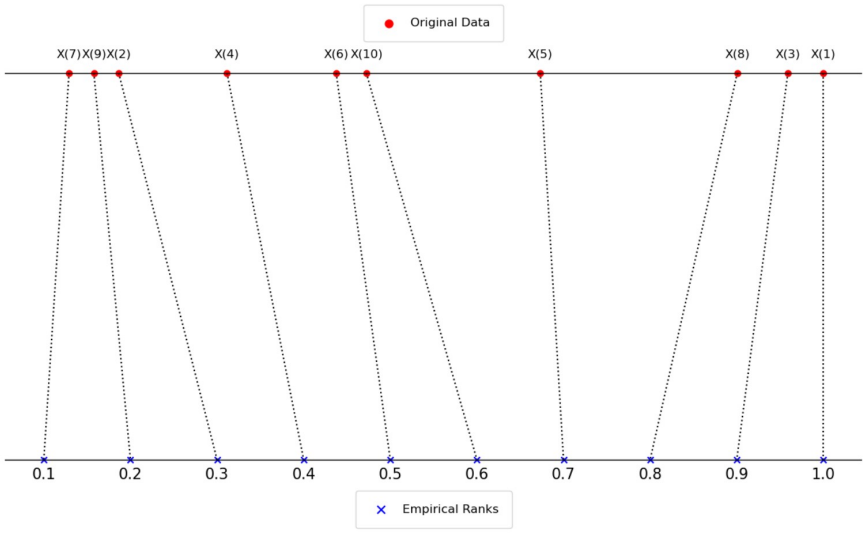


Figure 2: Univariate Ranks

Relationship between univariate data points and their corresponding ranks.

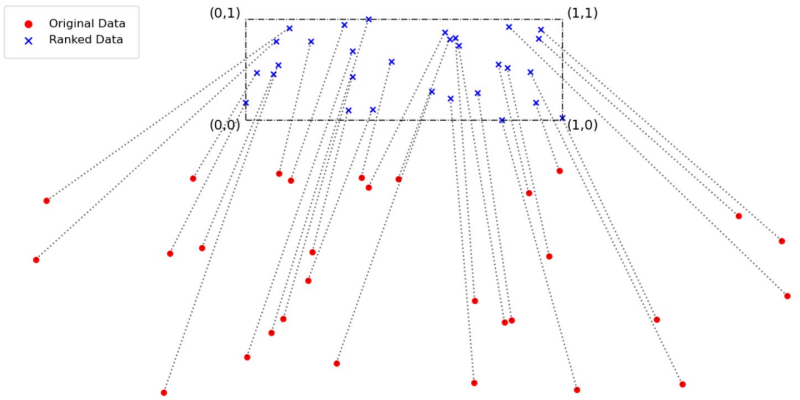


Figure 3: Bivariate Data Points and Their Ranks

Relationship between bivariate data points and their corresponding ranks. Data points and rank points are pseudo-random numbers generated using Uniform distribution, with rank points generated by $Uniform[0,1]^2$.

The above idea can be naturally generalized to multivariate case, through replacing the set of real numbers $\{1, 2, \ldots, n\}$ by the set of vectors in $\mathbb{R}^d$: $\{u_1, u_2, \ldots, u_n\} \subset [0,1]^d$. Now, the multivariate ranks is defined as

$$\widehat{R} = \text{argmin}_{\sigma=(\sigma(1),\ldots,\sigma(n))\in S_n} \sum_{i=1}^{n} ||X_i - u_{\sigma(i)}||^2 \tag{1}$$

where $||\cdot||^2$ is the Euclidean norm in $\mathbb{R}^d$. This can be viewed as an assignment problem, a topic that was discussed in Brenier (1991). The idea is illustrated above in Figure 3, and notice that central data points are mapped to points in the center of the rectangle, while marginal data points are mapped to points along the boundary of the rectangle.

The above optimization problem provides a distribution-free notion of multivariate ranks, while holding the desirable feature of computational efficiency (compared to the Tudey depth, which we will cover in the following discussion). Moreover, it can generate contours that capture non-convex feature better than the Tukey depth in certain cases, as illustrated in Figure 4.

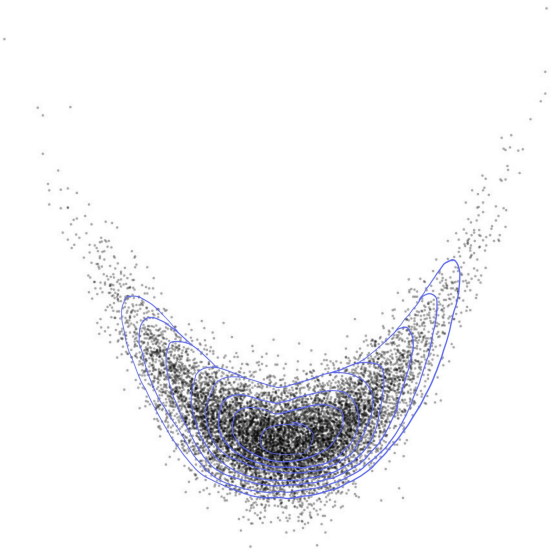


Figure 4: The Banana-shaped Distribution Example

The banana-shaped distribution and the corresponding optimal transport-based ranks contour.

In fact, the optimization problem is related to the optimal transport problem since it indeed transports the empirical distribution of X to the empirical distribution of u, which offers us a new perspective on defining multivariate ranks.

Here we give a brief introduction about the optimal transport theory. Taking the allocation problem as an example. A businessman runs several warehouses and factories. Suppose the total raw material supply of all warehouses equals the total demand of all factories. Our goal is to move all the resources from warehouses to factories, such that demand of all factories is successfully met, and the total transport cost is minimized. Now think of warehouses as our target distribution μ, and factories as a reference distribution ν. We aim to find a transport map between μ and ν that minimize the transport cost.

Optimal transport has found broad usage in various fields. See recent applications in domain adaptation from Fatras et al. (2021) and Chang et al. (2022), deep generative models from Rout et al. (2021) and Tong et al. (2023), statistical learning from Klink et al. (2022) and Montesuma et al. (2024), computer vision from Bonneel and Digne (2023) and Izquierdo and Civera (2024).

Traditionally, methods including solving differential equations (Benamou et al.

(2002)) and linear programming problems (Pele and Werman (2009)) were applied in handling the optimal transport problem. Peyré et al. (2019) showed that these algorithms have a complexity of $O(n^3 \log(n))$ when solving the problem. To reduce the computational cost of the optimal transport problem, a great deal of work dedicated in this topic these years. Zhang et al. (2021) indicates that two major techniques have been adopted to handle this problem. One is solving the regularized optimal transport problem, which reduce the computational cost to $O(n^2 \log(n))$. The other is estimating the optimal transport map through random or deterministic projections, which reduce the computational cost to $O(Kn \log(n))$, where K is the number of iterations until convergence. With these computational tools, optimal transport methods have been adopted in a wider range of applications.

3 Methodology

3.1 Preliminaries: Optimal Transport Theory

We first introduce several definitions for the following discussion.

Definition 3.1 (Borel probability measure). For measure space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ where $\mathcal{B}(\mathbb{R}^d)$ is the Borel σ-algebra, define

$$P : \mathcal{B}(\mathbb{R}^d) \to [0, 1]$$

as a Borel probability measure if it is a probability measure that satisfies non-negativity, unit measure of the whole sample space and σ-additivity.

Definition 3.2 ((Borel) measurable function). For measure space $(Y, \mathcal{F})$ and $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, function $F : Y \to \mathbb{R}$ is defined as a measurable function if

$$f^{-1}(B) \in \mathcal{F} \quad \text{for } \forall \text{ set } B \in \mathcal{B}(\mathbb{R})$$

where $f^{-1}(B) = \{y \in Y \mid f(y) \in B\}$.

Definition 3.3 (Absolutely continuous distribution). Suppose a random variable X whose probability measure is P. P is defined as an absolutely continuous distribution if there exists a function $f(x)$ such that

$$P_X(B) = \int_B f(x)dx \quad \text{for } \forall \text{ set } B \in \mathcal{B}(\mathbb{R}^d)$$

where $f(x)$ is referred to as the density function (PDF) of X.

Definition 3.4 (Subdifferential set). For convex function $f : \mathbb{R}^d \to \mathbb{R}$, the subdifferential set for f at point x is defined as

$$\partial f(x) = \{g \in \mathbb{R}^d \mid f(y) \geq f(x) + \langle y - x, g \rangle \text{ for } \forall\ y \in \mathbb{R}^d\}$$

Definition 3.5 (Legendre-Fenchel dual). For convex function $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$, the Legendre-Fenchel dual for f is defined as

$$f^*(y) = \sup_{x \in \mathbb{R}^d} (\langle y, x \rangle - f(x))$$

where $f^*(y)$ is also known as the conjugate of $f(x)$.

With these preliminaries, we outline the optimal transport (OT) theory in section 3.1, and OT-based definition of multivariate ranks and quantiles in section 3.2. Suppose μ and ν to be two Borel probability measure supported on $\mathcal{S} \subset \mathbb{R}^d$ and $\mathcal{Y} \subset \mathbb{R}^d$ respectively.

Definition 3.6 (Transport map). A measurable function $T : \mathcal{S} \to \mathcal{Y}$ is defined as a transport map from μ to ν if

$$\mu(T^{-1}(B)) = \nu(B) \quad \text{for } \forall\ \nu\text{-measurable set } B \in \mathcal{B}(\mathcal{Y})$$

where we often denote $\mu(T^{-1}(B))$ as $T\#\mu(B)$, and $T\#\mu$ is referred to as the push forward of μ by T. This process is denoted as $T\#\mu = \nu$.

An important problem in the OT theory is the Monge's OT problem (Monge (1781)), which is

$$\min_T \int_{\mathcal{S}} c(u, T(u)) d\mu(u) \quad \text{s.t. } T\#\mu = \nu \tag{2}$$

where T is a transport map from μ to ν, and $c : \mathcal{S} \times \mathcal{Y} \to \mathbb{R} \cup \{+\infty\}$ is the cost function. Generally speaking, we typically apply the Euclidean loss function as the cost function, therefore, the Monge's OT problem can be rewritten as

$$\min_T \int_{\mathcal{S}} ||u - T(u)||^2 d\mu(u) \quad \text{s.t. } T\#\mu = \nu \tag{3}$$

An essential result regarding the Monge's OT problem is the Brenier–McCann theorem.

Theorem 3.1 (Brenier–McCann theorem). Suppose μ and ν to be two Borel probability measures supported on $\mathbb{R}^d$, and suppose μ to be absolutely continuous. There exists a convex function $\psi : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ whose gradient $\nabla\psi : \mathbb{R}^d \to \mathbb{R}^d$ pushes μ forward to ν, and $\nabla\psi$ is unique for μ almost everywhere. Moreover, if μ and ν have finite second moments, $\nabla\psi$ uniquely minimizes the Monge's OT problem (2).

Another important theorem in convex optimization that will facilitate our following discussion is the Alexandrov theorem.

Theorem 3.2 (Alexandrov theorem). Suppose U to be an open subset of $\mathbb{R}^n$, and $f : U \to \mathbb{R}^n$ is a convex function, then f is differentiable almost everywhere.

With these preliminaries, we are now ready to define our ranks and quantiles.

3.2 OT-based Definition of Ranks and Quantiles

In this section, we introduce the definition of OT-based multivariate ranks and quantiles, both in population and sample perspective. Our discussion is based on the work of Ghosal and Sen (2022) and Deb and Sen (2023).

3.2.1 Population Ranks and Quantiles

Suppose μ to be a known absolutely continuous distribution on $\mathbb{R}^d$ with support $\mathcal{S}$, and specifically, we take μ to be Uniform$[0,1]^d$. Suppose random variable X with distribution ν is supported on $\mathcal{Y} \subset \mathbb{R}^d$.

Derive from Theorem 3.1, we can now define the rank map and the quantile function using the convex function $\psi : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ whose gradient $\nabla\psi : \mathbb{R}^d \to \mathbb{R}^d$, that is unique for μ almost everywhere, pushes μ forward to ν.

Definition 3.7 (Quantile function). The quantile function Q of ν w.r.t. μ is defined as the Q that pushes μ forward to ν.

$$Q : \mathcal{S} \to \mathcal{Y}$$
$$Q = \nabla\psi$$

Definition 3.8 (Rank map). The rank map R of ν w.r.t. μ is defined as the R that pushes ν forward to μ.

$$R : \mathcal{Y} \to \mathcal{S}$$
$$R = \nabla\psi^*$$

where $\psi^* : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ is the Legendre-Fenchel dual of ψ.

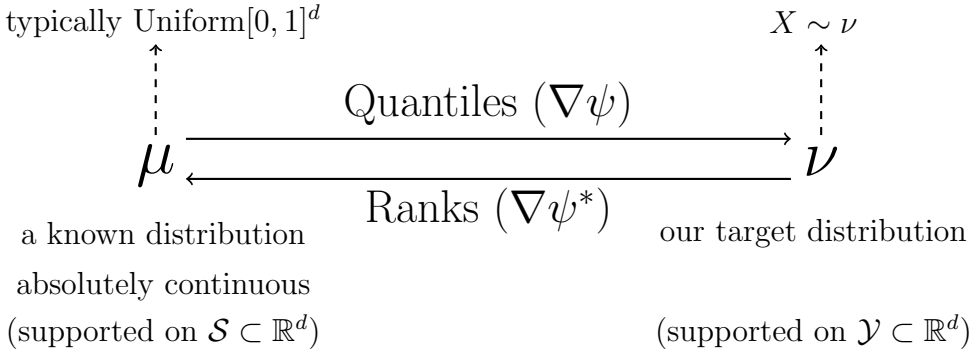


Figure 5: Main idea of the population ranks and quantiles

This shows the main idea of how to define OT-based multivariate ranks and quantiles: the transport map from μ to ν yields the quantiles of ν w.r.t. μ, while the transport map from ν to μ yields the ranks of ν w.r.t. μ.

From Theorem 3.1, we know that the quantile function $Q(u)$ (i.e. $\nabla\psi$) is unique w.r.t. μ for almost every u. When ψ is not differentiable at any point u, we can define $Q(u)$ to be any element in the subdifferential set $\partial\psi(u)$. Moreover, when μ and ν have finite second moment, $Q(u)$ can be expressed as

$$Q(u) = \text{argmin}_{T:T\#\mu=\nu}\, \mathbb{E}(||u - T(u)||^2)$$

just as the motivation that we discussed in section 2.2.

It has been discussed that $\psi^*(x) < \infty$ for any $x \in \mathcal{Y}$. Thus the convex function $\psi^*(x)$ is differentiable almost everywhere from Theorem 3.2, which leads to $R(x)$ is uniquely defined for almost every x. And when ψ^* is not differentiable at any point x, we can define $R(x)$ to be any element in the subdifferential set $\partial\psi^*(x)$, which we will discuss later.

If we suppose further v to be absolutely continuous, we have

$$\begin{aligned} R \circ Q(u) &= u \quad \text{for } u \text{ almost everywhere w.r.t. } \mu \\ Q \circ R(x) &= x \quad \text{for } x \text{ almost everywhere w.r.t. } \nu \end{aligned}$$

which implies that the quantile function Q and the rank map R are essential inverses (invertible almost everywhere) of each other. This aligns with the situation when $d = 1$.

3.2.2 Sample Ranks and Quantiles: Cell Decomposition

In this section, we follow the work of Ghosal and Sen (2022) to introduce the first definition and computation method of sample ranks and quantiles, which dependends on the concept of cell decomposition.

Given a random sample $\mathcal{X} = \{X_1, X_2, \ldots, X_n\}$ drawn from the distribution ν, we now define the sample ranks and quantiles. This is based on the empirical probability measure

$$\widehat{\nu}_n(A) = \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}(A) \quad \text{for } \forall \text{ set } A \subset \mathcal{B}(\mathbb{R}^d)$$

where $\delta_{X_i}(\cdot)$ is the indicator function

$$\delta_{X_i}(A) = \mathbf{1}_A(X_i) = \begin{cases} 0 & X_i \notin A \\ 1 & X_i \in A \end{cases}$$

Actually, $\widehat{\nu}_n$ is the empirical distribution of ν. Our idea is to replace the ν in the population setting with $\widehat{\nu}_n$.

Still derive from Theorem 3.1, we can now define the empirical rank map and the empirical quantile function using the convex function $\widehat{\psi}_n : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$.

Definition 3.9 (a) (Empirical quantile function). The empirical quantile function $\widehat{Q}_n$ of $\widehat{\nu}_n$ w.r.t. μ is defined as the $\widehat{Q}_n$ that pushes μ forward to $\widehat{\nu}_n$.

$$\begin{aligned} &\widehat{Q}_n : \mathcal{S} \to \mathcal{Y} \\ &\widehat{Q}_n = \nabla\widehat{\psi}_n \end{aligned}$$

Definition 3.10 (a) (Empirical rank map). The rank map $\widehat{R}_n$ of $\widehat{\nu}_n$ w.r.t. μ is defined as the $\widehat{R}_n$ that pushes $\widehat{\nu}_n$ forward to μ.

$$\widehat{R}_n : \mathcal{Y} \to \mathcal{S}$$
$$\widehat{R}_n = \nabla \widehat{\psi}_n^*$$

where $\widehat{\psi}_n^* : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ is the Legendre-Fenchel dual of $\widehat{\psi}_n$.

When μ and ν have finite second moment, $\widehat{Q}_n(u)$ can be expressed as

$$\begin{aligned}\widehat{Q}_n(u) &= \operatorname{argmin}_{T:T\#\mu=\widehat{\nu}_n} \mathbb{E}(||u - T(u)||^2) \\ &= \operatorname*{argmin}_T \int_{\mathcal{S}} ||u - T(u)||^2 d\mu(u) \quad \text{s.t.} \quad T\#\mu = \widehat{\nu}_n\end{aligned}$$

Just as we discussed in the population setting, $\widehat{Q}_n(u)$ (i.e. $\nabla\widehat{\psi}_n$) is unique w.r.t. μ for almost every u. And when $\widehat{\psi}_n$ is not differentiable at any point u, we can define $\widehat{Q}_n(u)$ to be any element in $\partial\widehat{\psi}_n(u)$. Since $\nabla\widehat{\psi}_n$ takes values only from $\mathcal{X}$, we have $\widehat{\psi}_n$ is piecewise linear, i.e. there exists $\widehat{h} = (\widehat{h}_1, \ldots, \widehat{h}_n)$ such that $\widehat{\psi}_n$ can be written as

$$\widehat{\psi}_n(u) = \begin{cases} \max\{u^T X_i + \widehat{h}_i\} & u \in \mathcal{S} \\ +\infty & u \notin \mathcal{S} \end{cases} \tag{4}$$

where $\widehat{h}$ can be uniquely calculated by solving a convex optimization problem, which we will discuss later. Given the above formulation of $\widehat{\psi}_n$, we have the subdifferential of it is expressed as a convex hull

$$\partial\widehat{\psi}_n(u) = \operatorname{Conv}(\{X_i \mid \langle u, X_i\rangle + \widehat{h}_i = \widehat{\psi}_n(u)\}) \quad \text{for } \forall\ u \in \mathcal{S}$$

In fact, $\widehat{Q}_n = \nabla\widehat{\psi}_n$ leads to a cell decomposition of $\mathcal{S}$, where each cell is convex and defined as

$$W_i(\widehat{h}) = \{u \in \mathcal{S} \mid \nabla\widehat{\psi}_n(u) = X_i\} \tag{5}$$

Every $W_i(\widehat{h})$ has a measure of $\frac{1}{n}$ w.r.t. μ, and $\bigcup_{i=1}^n W_i(\widehat{h}) \subset \mathcal{S}$.

Given this cell decomposition, we can use the following theorem to find the empirical rank map of any interior point (here the interior refers to the interior of convex hulls of several data points)

Theorem 3.3 (Empirical rank map-finding). For $x \in \mathbb{R}^d$, suppose for $\{i_1, i_2, \ldots, i_{d+1}\} \subset \{1, 2, \ldots, n\}$ we have

(i) $x \in \operatorname{Int}(\operatorname{Conv}(X_{i_1}, X_{i_2}, \ldots, X_{i_{d+1}}))$, where Int denotes the interior of a set

(ii) there exists a unique $u \in \mathcal{S}$ such that $u = \mathrm{Cl}(W_{i_1}(\widehat{h})) \cap \cdots \cap \mathrm{Cl}(W_{i_{d+1}}(\widehat{h}))$, where Cl denotes the closure of a set

Then u is the unique point in $\mathcal{S}$ such that $x \in \partial\widehat{\psi}_n(u)$, and $\widehat{R}_n(x) = \partial\widehat{\psi}_n^*(x) = u$.

However, cases are for boundary points, and specifically our data points, we can define $\widehat{R}_n(X_i)$ to be any point in the closure of $W_i(\widehat{h})$, specifically, a random point drawn from the distribution Uniform$(W_i(\widehat{h}))$. It was proved that $\widehat{R}_n(X_i) \sim \mu$, which aligns with the population result. Moreover, it also implies that the multivariate ranks are marginally distribution-free.

Based on the cell decomposition in section 3.2.2, we can now calculate the sample ranks and quantiles by solving convex optimization problems via Newton methods. The theory of power diagram is also involved in the discussion: given our random sample $\mathcal{X}$ and a vector $w = (w_1, w_2, \ldots, w_n) \in \mathbb{R}^n$, the power diagram of $(\mathcal{X}, w)$ decomposes $\mathcal{S}$ into n cells with each X_i in an unique cell, and the volume of each cell is $\frac{1}{n}$. Each cell is defined as

$$\mathrm{Pow}_{\mathcal{X}}^{w}(i) = \{u \in \mathcal{S} : ||u - X_i||^2 - w_i \leq ||u - X_j||^2 - w_j \text{ for } \forall\ j \neq i\}$$

To find the w that satisfies $\mu(\mathrm{Pow}_{\mathcal{X}}^{w}(i)) = \frac{1}{n}$, we solve

$$\widehat{w} = \underset{w \in \mathbb{R}^n}{\mathrm{argmax}} \sum_{i=1}^{n} \left[\frac{w_i}{n} + \int_{\mathrm{Pow}_{\mathcal{X}}^{w}(i)} (||u - X_i||^2 - w_i) d\mu(u) \right] \tag{6}$$

Now we define the power map

Definition 3.11 (Power map). Power map is defined as

$$T_{\mathcal{X}}^{w} : \mathcal{S} \to \mathcal{X}$$
$$T_{\mathcal{X}}^{w}(u) = X_i \text{ if } u \in \mathrm{Pow}_{\mathcal{X}}^{w}(i)$$

With the power map, we have our sample quantile

$$\widehat{Q}_n = T_{\mathcal{X}}^{\widehat{w}}$$

where $\widehat{w}$ is calculated in (5).

If we define $\widehat{h}_i = \frac{1}{2}(\widehat{w}_i - ||X_i||^2)$, we now have the $\widehat{\psi}_n(u)$ from (4). Thus we calculate the $W_i(\widehat{h})$ in (5) by

$$W_i(\widehat{h}) = \underset{u \in \mathcal{S}}{\mathrm{argmax}} \{\langle u, X_i \rangle - \widehat{\psi}_n(u)\}$$

Then we have the sample rank $\widehat{R}_n(X_i)$ that is randomly chosen from $W_i(\widehat{h})$.

3.2.3 Sample Ranks and Quantiles: A Quasi-Monte Carlo Method

Although the above discussion in section 3.2.2 is already enough for definition and calculation, we can still make life easier by introducing another approach, a quasi-Monte Carlo method, to our work.

Recall that in section 2.2, we introduced the motivation of generalizing univariate ranks into multivariate cases in sample-wise by viewing the rank-finding process as an assignment problem, which is formulated mathematically in (1). Following our population definition in section 3.1, we can now further elaborate on this idea.

The principle lies in that, instead of pushing forward our random sample to the continuous distribution $\mu = \text{Uniform}[0,1]^d$, we directly discretize it into a set of vectors in $\mathbb{R}^d$: $\{u_1, u_2, \ldots, u_n\} \subset [0,1]^d$, where we sample $\{u_1, u_2, \ldots, u_n\}$ from the $\text{Uniform}[0,1]^d$ using a quasi-Monte Carlo method. And we specifically advocate the use of Halton sequence.

One may question why quasi-Monte Carlo method is used. The advantage lies in the following points: (i) compared with simply random sampling, quasi-Monte Carlo sequence is deterministic; (ii) compared with simply sampling from grid, Quasi-Monte Carlo sequence can be constructed for every n and d; (iii) when the sample size n changes, we do not have to completely sample again; (iv) since quasi-Monte Carlo sequence is a low-discrepancy sequence, it can provide greater uniformity.

Halton sequence can be generated by the `Python` function `Halton()` in the `SciPy` repository. The basic idea of sampling a Halton sequence is

(1) Selecting different prime numbers in each dimension as a base number p.

(2) Convert the data index i into base p representation. Then reverse its fractional digits and map it to the [0,1] interval.

(3) Compute Halton values independently for each dimension and combine them into a d-dimensional sample.

To showcase different ways of sampling, we generate samples using three methods and are shown below.

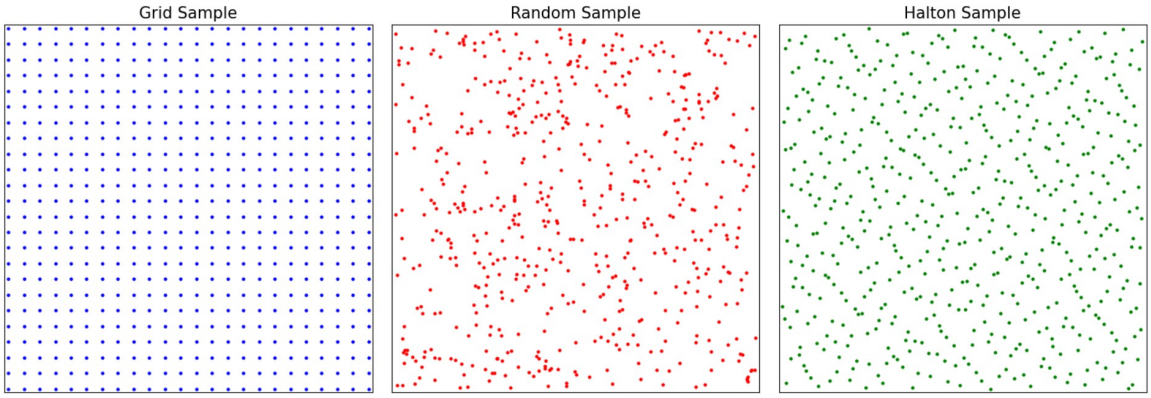


Figure 6: Comparison Between Different Sampling Methods

We draw $n = 24 \times 24 = 576$ points using grid sampling, simple random sampling and Halton sampling. Notice that (i) when n is not a perfect square, we cannot use the grid sampling; (ii) comparing simple random sampling with Halton sampling, it can be intuitively concluded that points are far more uniformly distributed using the latter one.

Now we are able to define the empirical ranks and quantiles using the Halton sequence $H_n = \{h_1, h_2, \ldots, h_n\} \subset [0,1]^d$.

Definition 3.10 (b) (Empirical rank). The rank $\widehat{R}_n$ is defined as

$$\widehat{R}_n(X_i) = h_{\sigma^*(i)}$$

where $\sigma^* = (\sigma^*(1), \ldots, \sigma^*(n))$ is calculated from

$$\sigma^* = \text{argmin}_{\sigma=(\sigma(1),\ldots,\sigma(n))\in S_n} \sum_{i=1}^{n} ||X_i - h_{\sigma(i)}||^2$$

and S_n is the set of all permutation of $\{1, 2, \ldots, n\}$.

Definition 3.9 (b) (Empirical quantile). The empirical quantile function $\widehat{Q}_n$ is defined as

$$\widehat{Q}_n(u) = \underset{X_i}{\text{argmin}} \, ||\widehat{R}_n(X_i) - u||$$

where $||\cdot||$ is the $L1$ norm in $\mathbb{R}^d$.

Following the definitions, we outline basic concepts of the assignment problem and related algorithms to handle the problem. Recall the warehouses and factories example we introduced in section 2.2, now we require that each factory receives all resources from one and only one warehouse. Our goal is still to minimize the total assignment cost. Denote the cost matrix as $C = [c_{ij}]_{n\times n}$, where c_{ij} is the assignment cost from the ith warehouse to the jth factory. In our specific case, $c_{ij} = ||X_i - h_j||^2$. The assignment

problem can be mathematically formulated as

$$\min \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} w_{ij}$$
$$\text{s.t.} \quad \sum_{i=1}^{n} w_{ij} = 1, \forall j; \ \sum_{j=1}^{n} w_{ij} = 1, \forall i; \ w_{ij} \in \{0, 1\}$$

The assignment problem has been studied extensively in the combinatorial optimization literature (see Munkres (1957) and Bertsekas (1988)) and various LP algorithms were proposed, among which is the most famous Hungarian algorithm (see Jonker and Volgenant (1988)) that has worst case computational complexity of $O(n^3)$.

Algorithm 1 Hungarian Algorithm

Require: Cost matrix $C = [c_{ij}]_{n \times n}$
Ensure: Optimal assignment matrix $W = [w_{ij}]_{n \times n}$
Step 1: Row Reduction
for $i = 1$ to n **do**
    $c_{i\cdot} = c_{i\cdot} - \min(c_{i\cdot})$
Step 2: Column Reduction
for $j = 1$ to n **do**
    $c_{\cdot j} = c_{\cdot j} - \min(c_{\cdot j})$
Step 3: Cover all zeros with a minimum number of lines
repeat
    Identify all zero positions in C
    Cover all zeros using the minimum number of horizontal and vertical lines
    if number of lines == n **then**
        Go to Step 5
    else
        Go to Step 4
until number of lines == n
Step 4: Adjust the Matrix C
Find the minimum uncovered element δ
for each c_{ij} **do**
    if element is uncovered **then**
        $c_{ij} = c_{ij} - \delta$
    else if element is covered twice **then**
        $c_{ij} = c_{ij} + \delta$
Step 5: Find the Optimal Assignment
Select independent zeros to form a unique assignment
Return assignment matrix W

The Hungarian algorithm can be realized by the `Python` function `linear_sum_assignment()` in the `SciPy` repository.

3.3 Multivariate Nonparametric Testing Using OT-based Ranks

With our previously defined OT-based multivariate ranks, we are now able to construct nonparametric testing in multivariate cases. The general principle is straightforward. If $T_n = T(X_1, \ldots, X_n)$ is the test statistic employed in any current approach, we can then construct the multivariate distribution-free counterpart of it by replacing X_is with their ranks $\widehat{R}(X_i)$s to get the new test statistic

$$T_n^{'} = T(\widehat{R}(X_i), \ldots, \widehat{R}(X_n))$$

It is evident that this provides a general framework for constructing diverse distribution-free testing procedures.

In fact, since the proposal of OT-based multivariate ranks, a large body of literature has investigated the topic of nonparametric testing, with two-sample goodness-of-fit testing and mutual independence testing being the primary focuses of discussion. Given two random samples $X_1, \ldots, X_n$ in $\mathbb{R}^d$ with distribution $X \sim \nu_X$, and $Y_1, \ldots, Y_m$ in $\mathbb{R}^d$ with distribution $Y \sim \nu_Y$, the two sample goodness-of-fit test aims to test

$$H_0 : \nu_X = \nu_Y \quad \text{versus} \quad H_1 : \nu_X \neq \nu_Y$$

Suppose the random vector $X = (X^{(1)}, \ldots, X^{(k)})$ in $\mathbb{R}^d$ with $X \sim \nu$, and $X^{(i)} \sim \nu_i$ in $\mathbb{R}^{d_i}$ for $i = 1, \ldots, k$, where $\sum_{i=1}^k d_i = d$. We have random sample $X_1, \ldots, X_n$ that is drawn from ν. The mutual independence test (w.r.t. ν_is) aims to test

$$H_0 : \nu = \nu_1 \otimes \cdots \otimes \nu_k \quad \text{versus} \quad H_1 : \nu \neq \nu_1 \otimes \cdots \otimes \nu_k$$

Now we carry out a literature review to motivate our multivariate nonparametric tests.

One of the goodness-of-fit tests was proposed by Deb et al. (2021). They introduced the rank Hotelling T^2 statistic

$$T_{m,n}^{\nu} = \frac{mn}{m+n} \left\| \frac{1}{n} \sum_{i=1}^{n} \widehat{R}_{m,n}(X_i) - \frac{1}{m} \sum_{j=1}^{m} \widehat{R}_{m,n}(Y_i) \right\|^2$$

where $\widehat{R}_{m,n}(\cdot)$ is the empirical rank function of the pooled sample $\{X_1, \ldots, X_n, Y_1, \ldots, Y_m\}$. This test statistic can be viewed as the rank version of the Hotelling T^2 test

$$T_{m,n} = \frac{mn}{m+n} (\bar{X} - \bar{Y})^T S_{m,n}^{-1} (\bar{X} - \bar{Y})$$

where $\bar{X}$ and $\bar{Y}$ are sample means and $S_{m,n}$ is the variance covariance matrix of the

pooled sample

$$S_{m,n} = \frac{1}{m+n-2}\left(\sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})^T + \sum_{j=1}^{m}(Y_i - \bar{Y})(Y_i - \bar{Y})^T\right)$$

since $T^{\nu}_{m,n}$ is constructed by replacing observations with their empirical pooled ranks. Moreover, when $d = 1$, they showed that the statistic $T^{\nu}_{m,n}$ is equivalent to the Wilcoxon rank-sum statistic.

Another goodness-of-fit test was proposed by Deb and Sen (2023). They introduced the rank energy statistic

$$\begin{aligned} RE^2_{m,n} =& \frac{2}{mn}\sum_{j=1}^{m}\sum_{i=1}^{n}\left\|\widehat{R}_{m,n}(X_i) - \widehat{R}_{m,n}(Y_j)\right\|^2 \\ &- \frac{1}{n^2}\sum_{i,j=1}^{n}\left\|\widehat{R}_{m,n}(X_i) - \widehat{R}_{m,n}(X_j)\right\|^2 \\ &- \frac{1}{m^2}\sum_{i,j=1}^{m}\left\|\widehat{R}_{m,n}(Y_i) - \widehat{R}_{m,n}(Y_j)\right\|^2 \end{aligned}$$

where $\widehat{R}_{m,n}(\cdot)$ is the empirical rank function of the pooled sample $\{X_1, \ldots, X_n, Y_1, \ldots, Y_m\}$. When $d = 1$, recall the Cramér-von Mises (CvM) statistic

$$CM_{m,n} = \int (F^X_n(t) - G^Y_m(t))^2 dH^{X,Y}_{m,n}(t)$$

where F^X_n, G^Y_m and $H^{X,Y}_{m,n}$ denote the empirical CDF of X, Y and the pooled sample. They showed that

$$\frac{1}{2}RE^2_{m,n} = CM_{m,n}$$

Deb (2022) and Masud et al. (2021) also proposed test based on the concept of rank energy.

Some other goodness-of-fit tests include tests inspired from the Wasserstein distance by Hallin et al. (2021b) and González-Delgado et al. (2023), the test derived from the CvM test by Ghosal and Sen (2022), tests derived form the Wilcoxon rank sum test by Huang and Sen (2023) and Betken et al. (2024), and the test measuring the distance between characteristic functions (with r.v. replaced by rank) by Deb and Sen (2023). As for testing for independence, Deb and Sen (2023) proposed their test by measuring distance between joint ranks and marginal ranks, Niu and Bhattacharya (2022) and Deb (2022) developed tests by quantifying distance between joint characteristic functions, and Shi et al. (2025) proposed their kendall τ and spearman ρ tests using ranks.

4 Proposed Tests

In this section, we propose three new multivariate tests, namely the rank minimal spanning tree test, the rank Anderson Darling test, and the multivariate Wald–Wolfowitz runs test. We will discuss the motivation behind each test, the mathematical articulation (especially the test statistic) and the pseudo-code of each one, before explaining the calculation of each test's p value.

4.1 The Rank Minimal Spanning Tree Test

When taking a closer look at the proposed rank Hotelling T^2 test by Deb et al. (2021), we can view it as the generalization of the univariate T test through two pathways. One is to first view the univariate Wilcoxon rank sum test as the rank version of the univariate T test, and then extend it to its multivariate version; the other, instead, is to initially generalize the univariate T test to its multivariate version, before deriving its rank version which is the rank Hotelling T^2 test. Permutation is recommended to get the p values of the test in practice.

Before moving forward, we would like to first discuss why permutation works (the reason why we can obtain a valid p value through permutation). This can be explained by the so-called exchangeability. For our goodness-of-fit testing, we aim to test if $\nu_X = \nu_Y$ using the test statistic $T = T(X_1, \ldots, X_n, Y_1, \ldots, Y_m)$. Under the null hypothesis, the pooled sample

$$\begin{aligned}(X, Y) &= (X_1, \ldots, X_n, Y_1, \ldots, Y_m) \\ &= (Z_1, \ldots, Z_{m+n})\end{aligned}$$

is an i.i.d. sample, which leads to the fact that after the permutation, the joint distribution of $(X', Y') = (Z_{i_1}, \ldots, Z_{i_{m+n}})$ remains the same as the joint distribution of (X, Y). Therefore, the values T' obtained from permutations can be used to construct an empirical distribution that approximates the null distribution of T. This means we can compute the p value by calculating the proportion of T' values that are as extreme as or more extreme than the observed T.

Now, upon closer consideration of the rank Hotelling T^2 test, we can tell that it has several drawbacks. The most significant one of is that it focuses solely on differences in the overall mean of the distributions, rather than evaluating discrepancies across the full range of data points. As a result, it tends to overlook many details in the data. Moreover, although the rank Hotelling T^2 test is robust to non-normality due to the rank transformation, the statistical foundation of it remains largely linear-algebraic. This may reduce its ability to detect highly non-linear relationships or

complex dependencies in high-dimensional settings.

To handle the drawbacks of the rank Hotelling T^2 test, we propose the Rank Minimal Spanning Tree test (Rank MST test). To begin with, the principle of using the minimal spanning tree in statistical testing is: for two random samples, we construct the minimal spanning tree (the tree that connects all points with the minimum total sum of edge weights) of the pooled sample. The test statistic is defined as the amount of edges that connects points from different samples. If the two samples are generated from the same distribution, their data points tend to be nested within each other in the multivariate space, resulting in more edges in the minimal spanning tree connecting points from different samples. In contrast, if the two samples come from distinct distributions, their points tend to be separated in space, leading to fewer between-sample edges in the minimal spanning tree. Following this idea, the Rank MST test adopts the same procedure as the MST test, but computes distances between corresponding data points in the rank space.

Suppose we have two random samples $X_1, \ldots, X_n$ in $\mathbb{R}^d$ with distribution $X \sim \nu_X$, and $Y_1, \ldots, Y_m$ in $\mathbb{R}^d$ with distribution $Y \sim \nu_Y$. We aim to test

$$H_0 : \mu_{\nu_X} = \mu_{\nu_Y} \quad \text{versus} \quad H_1 : \nu_X \neq \nu_Y$$

Pool the two samples, and apply Algorithm 1 to get the OT-based ranks of the pooled sample $\{X_1, \ldots, X_n, Y_1, \ldots, Y_m\}$. Calculate the Euclidean distance of every two points' ranks in the pooled sample, and thus construct the minimal spanning tree (using the Prim algorithm or Kruskal algorithm). The test statistic is

$$\#\text{ Cross Sample Edges}$$

where $\#A$ refers to the number of A. P value of the test is usually obtained from permutation. The following shows the pseudocode of the proposed test.

Algorithm 2 Rank Minimal Spanning Tree (Rank MST) Test

Require: Two samples $X = \{X_1, \ldots, X_n\} \subset \mathbb{R}^d$ and $Y = \{Y_1, \ldots, Y_m\} \subset \mathbb{R}^d$; Number of permutations B.
Ensure: The p-value of the test.
Pool samples: $Z \leftarrow \{X_1, \ldots, X_n, Y_1, \ldots, Y_m\}$. Let $N \leftarrow n + m$.
Create label vector L of length N, where $L_i = 0$ if $Z_i \in X$ and $L_i = 1$ if $Z_i \in Y$.
Compute OT-based ranks for the pooled sample using the Hungarian algorithm: $R \leftarrow$ OT-RANK(Z).
procedure COMPUTESTATISTIC(R, L)
    Construct MST using the Kruskal algorithm on the points in R, where edge weights are Euclidean distances.
    edges $\leftarrow 0$.
    for each edge (i, j) in the MST **do**
        if $L_i \neq L_j$ **then**
            edges $\leftarrow$ edges $+ 1$.
    return edges.
$\text{stat}_{\text{obs}} \leftarrow$ COMPUTESTATISTIC(R, L).
    extreme_count $\leftarrow 0$.
for $b = 1$ to B **do**
    Permute the label vector L to get a shuffled vector L'.
    $\text{stat}_{\text{perm}} \leftarrow$ COMPUTESTATISTIC(R, L').
    if $\text{stat}_{\text{perm}} \leq \text{stat}_{\text{obs}}$ **then**
        extreme_count $\leftarrow$ extreme_count $+ 1$.
$p_{\text{value}} \leftarrow (\text{extreme_count} + 1)/(B + 1)$.
return p_{value}.

The computation complexity of the algorithm is $O(B \cdot n^3)$ where B is the times of permutation. Compared with the MST test, the Rank MST test relies on OT–based ranks to better capture the underlying geometric and dependency structure of multivariate data. It does not depend on any distributional assumptions, thus is not affected by variations in scale and distributional form across different dimensions of the original data. Instead of ranking each variable separately, these ranks take the full joint distribution into account, so as to preserve more information about how the variables relate to each other. When these ranks are used to build the minimal spanning tree, the test becomes more sensitive to subtle and complex differences between distributions (including changes in shape and nonlinear structure, etc) that might be missed by merely caring about average rank differences. The structure of the tree can also be used to reveal these differences visually (although only practical in $\mathbb{R}^2$ or $\mathbb{R}^3$), often showing clear group separations or distinctive connection patterns in the rank-transformed space. Moreover, the Rank MST test can also be used in testing for goodness-of-fitness in multiple sample cases.

4.2 The Rank Anderson Darling Test

Recall the rank energy statistic by Deb and Sen (2023), which was proved to be the multivariate version of the two sample CvM statistic (equal up to a constant multiplier). When we look at the two sample Anderson Darling test (AD test) in $\mathbb{R}^1$, whose test statistic is

$$AD_{m,n} = \int \frac{(F_n^X(t) - G_m^Y(t))^2}{H_{m,n}^{X,Y}(t)(1 - H_{m,n}^{X,Y}(t))} dH_{m,n}^{X,Y}(t)$$

we can readily observe that the difference between it and the CvM statistic lies in the fact that the former is divided by a certain term, which highlights a key feature of the AD test: it places more weight on observations in the tails of the distribution. Since the AD test gives more emphasis to differences in the distribution tails, it is well-suited for detecting discrepancies involving extreme values or variations in tail heaviness.

The above discussion motivates us to generalize the univariate two sample AD test into its multivariate counterpart via OT-based rank transformation. We propose the Rank Anderson Darling test (Rank AD test). The hypothesis is still

$$H_0 : \mu_{\nu_X} = \mu_{\nu_Y} \quad \text{versus} \quad H_1 : \nu_X \neq \nu_Y$$

and the test statistic is now

$$\begin{aligned} RAD_{m,n}^2 =& \frac{2}{mn} \sum_{j=1}^{m} \sum_{i=1}^{n} \frac{\left\| \widehat{R}_{m,n}(X_i) - \widehat{R}_{m,n}(Y_j) \right\|^2}{w(X_i) w(Y_j)} \\ & - \frac{1}{n^2} \sum_{i,j=1}^{n} \frac{\left\| \widehat{R}_{m,n}(X_i) - \widehat{R}_{m,n}(X_j) \right\|^2}{w(X_i) w(X_j)} \\ & - \frac{1}{m^2} \sum_{i,j=1}^{m} \frac{\left\| \widehat{R}_{m,n}(Y_i) - \widehat{R}_{m,n}(Y_j) \right\|^2}{w(Y_i) w(Y_j)} \end{aligned}$$

where for any $Z \in \mathbb{R}^d$,

$$w(Z) = \prod_{k=1}^{d} \widehat{R}_{m,n}(Z^{(k)}) \cdot (1 - \widehat{R}_{m,n}(Z^{(k)}))$$

and $Z^{(k)}$ is the k-th component of Z. P value of the test can be obtained from permutation.

Under the null hypothesis, the pooled ranks $\widehat{R}_{m,n}(\cdot)$ are uniformly distributed and the weight term $w(\cdot)$ is a known function of these uniform ranks. Therefore, the test

statistic of the Rank AD test does not rely on the distribution of the original data, which is a desirable property.

The following shows the pseudocode of the proposed test.

Algorithm 3 Rank Anderson-Darling (Rank AD) Test

Require: Two samples $\{X_1, \ldots, X_n\} \subset \mathbb{R}^d$, $\{Y_1, \ldots, Y_m\} \subset \mathbb{R}^d$
Ensure: The p-value of the test.

Combine all samples into $Z = \{X_1, \ldots, X_n, Y_1, \ldots, Y_m\}$
Compute empirical OT ranks: R = RankTransform(Z)
for each $z \in Z$ **do**
    $w(z) \leftarrow 1$
    for $k = 1$ to d **do**
        $r \leftarrow R_k(z)$
        $w(z) \leftarrow w(z) \cdot r \cdot (1 - r)$
$T_1 \leftarrow 0$
for $i = 1$ to n **do**
    for $j = 1$ to m **do**
        $d \leftarrow \|R(X_i) - R(Y_j)\|^2$
        $T_1 \leftarrow T_1 + d/(w(X_i) \cdot w(Y_j))$
$T_1 \leftarrow (2/(m \cdot n)) \cdot T_1$
$T_2 \leftarrow 0$
for $i = 1$ to n **do**
    for $j = 1$ to n **do**
        $d \leftarrow \|R(X_i) - R(X_j)\|^2$
        $T_2 \leftarrow T_2 + d/(w(X_i) \cdot w(X_j))$
$T_2 \leftarrow T_2/n^2$
$T_3 \leftarrow 0$
for $i = 1$ to m **do**
    for $j = 1$ to m **do**
        $d \leftarrow \|R(Y_i) - R(Y_j)\|^2$
        $T_3 \leftarrow T_3 + d/(w(Y_i) \cdot w(Y_j))$
$T_3 \leftarrow T_3/m^2$
`RAD` $\leftarrow T_1 - T_2 - T_3$
p value $\leftarrow$ PermutationTest(X, Y, RAD)
return RAD, p value

4.3 The Multivariate Wald–Wolfowitz Runs Test

The Wald–Wolfowitz runs test (Runs test) is also an univariate nonparametric test based on the notion of ranks. Suppose we have two random samples $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ in $\mathbb{R}$. In order to test if the two samples are generated from the same distribution, we first pool the two samples to get $\{X_1, \ldots, X_n, Y_1, \ldots, Y_m\}$. We order each number from the pooled sample to get the ordered pooled sample and count the number of runs (# runs) in it, where a "run" refers to a sequence of numbers that belong

to the same original sample: $\{X_i\}$ or $\{Y_j\}$ (for instance, the set $\{X, X, Y, X, X, X, Y, Y\}$ contains four runs, i.e. $\{X, X\}$, $\{Y\}$, $\{X, X, X\}$ and $\{Y, Y\}$). If $\{X_i\}$ and $\{Y_j\}$ are generated from the sample distribution, numbers tend to be randomly distributed in the pooled sample, hence # runs tend to be more. The idea of the Runs test is pretty similar to that of the MST test.

To generalize the test into its multivariate counterpart through our notion of ranks, we first carry out the rank transformation on the pooled sample to get the pooled ranks set

$$\{\widehat{R}_{m,n}(X_1), \ldots, \widehat{R}_{m,n}(X_n), \ldots, \widehat{R}_{m,n}(Y_1), \ldots, \widehat{R}_{m,n}(Y_m)\}$$

However, we cannot order the pooled ranks in $\mathbb{R}^d$. To address this, we can choose from (but not limited to) the following methods

(1) Principal component analysis (PCA): Project the data in the pooled ranks set to a certain direction in $\mathbb{R}$, and then apply the univariate Runs test on the dimension-reduced pooled ranks set.

(2) Clustering: Suppose we cluster the pooled ranks sample into k clusters. Derive from the idea of the univariate Runs test, if the null hypothesis holds, ranks from both samples tend to distribute randomly in most of the clusters. Hence we can check the randomness of ranks from different samples in each cluster and construct our test statistic (for example, if 40 out of 50 clusters are considered to contain both samples distributed relatively evenly, then a permutation test can be performed based on this, and it is highly likely that we will fail to reject the null hypothesis).

(3) Hilbert sorting: Hilbert sorting is a technique that uses the Hilbert space-filling curve to sort high-dimensional data, where the Hilbert curve is a continuous, space-filling curve that maps points from a high-dimensional space to a one-dimensional space while preserving spatial locality as much as possible. The idea of using Hilbert sorting in out test is the same as using PCA.

If test statistics are properly chosen, the above methods are expected to be distribution-free. However, the dimension reduction process may lead to information loss, which is problematic when data dimension becomes high.

5 Simulation Study for the Rank MST Test

In this section, we generate several samples from multiple different distributions. The main purpose of this section is to showcase the performance, including p values, powers and computation time, of our proposed methods using the Rank MST test only (due to time and page limitations of the dissertation), as an example. We will carry out our simulation study in two representative scenarios: $\mathbb{R}^2$ and some intermediate dimensions.

Before moving on to the simulation, we start by discussing the reference distribution. In previous sections, the reference distribution was set as Unif$[0,1]^d$ (sampled from the Halton sequence in practice). When we think about this, we do not have to merely rely on Unif$[0,1]^d$. We can also choose the uniform unit ball in $\mathbb{R}^d$ (Ball$[0,1]^d$) as our reference distribution. In fact, the reason for doing this is that in our two-sample testing, when the original distribution is Gaussian or t (which is normal-like with heavier tails), Ball$[0,1]^d$ preserves the spatial structure of the data better than Unif$[0,1]^d$, so as to improve the effectiveness and robustness of our method. To clarify, when both samples are generated from normal-like distributions, we take Ball$[0,1]^d$ as our reference distribution, while we take Unif$[0,1]^d$ when at least one sample is not normal-like. Number of permutations are set as 1000, and we repeat each test for 50 times when sample size is $m = n = 200$ to get an average p value, while due to time limitation, we only repeat each test for 10 times when sample size is $m = n = 1000$. We compare the performance of our Rank MST test with that of the Energy Distance test, which is the non-rank version of the rank energy statistic that we have mentioned in section 3.3, the test statistic of which is

$$ED^2_{m,n} = \frac{2}{mn}\sum_{j=1}^{m}\sum_{i=1}^{n}\|X_i - Y_j\|^2 - \frac{1}{n^2}\sum_{i,j=1}^{n}\|X_i - X_j\|^2 - \frac{1}{m^2}\sum_{i,j=1}^{m}\|Y_i - Y_j\|^2$$

The reason why it serves as our benchmark is because it is a widely used, nonparametric test for two-sample goodness-of-fitness testing. Moreover, both Energy Distance and our Rank MST method are based on pairwise distances, which makes the comparison fair and meaningful. Moreover, Energy Distance serves as a strong baseline, and outperforming it in challenging settings highlights the strength of our method.

5.1 Testing in $\mathbb{R}^2$

5.1.1 Simulation When Components Are Independent

We start with the easiest case in $\mathbb{R}^2$. We first assume the two components in each r.v. are independent from each other. We generate samples from the following distributions

(a1) Norm(0, 0): X $\sim$ N($\boldsymbol{\mu}_{a1}$, $3\boldsymbol{I}_2$) where $\boldsymbol{\mu}_{a1} = [0,0]^{\mathrm{T}}$.

(a2) Norm(-2, -2): $X \sim N(\boldsymbol{\mu}_{a2}, 3\boldsymbol{I}_2)$ where $\boldsymbol{\mu}_{a2} = [-2, -2]^T$.

(a3) Unif(0, 6): $X \sim \text{Unif}[0, 6]^2$.

(a4) T(0, 2): $X \sim T(\boldsymbol{\mu}_{a4}, \boldsymbol{I}_2, \nu)$ where $\boldsymbol{\mu}_{a4} = [0, 2]^T$ and $\nu = 3$ is the degree of freedom.

(a5) T(1, 1): $X \sim T(\boldsymbol{\mu}_{a5}, \boldsymbol{I}_2, \nu)$ where $\boldsymbol{\mu}_{a5} = [1, 1]^T$.

It is easily seen that the covariance matrix from (a1) to (a5) are the same. The testing results are listed as follows

GOF Testing/Methods	**Rank MST**	**Energy Distance**
Norm(0, 0) & Norm(0, 0)	0.5413	0.4626
Norm(0, 0) & Norm(-2, -2)	0.0010	0.0010
Norm(0, 0) & Unif(0, 6)	0.0010	0.0010
Norm(0, 0) & T(0, 2)	0.0010	0.0010
Norm(0, 0) & T(1, 1)	0.0010	0.0010
T(0, 2) & T(1, 1)	0.0010	0.0010
Unif(0, 6) & T(1, 1)	0.0010	0.0010

Table 1: P Values When Components Are Independent ($m = n = 200$)

We can tell from the result that, both methods correctly distinguish the diffference between each two samples under the significance level of 0.05, and fail to reject the null hypothesis when the two samples come from the same distribution.

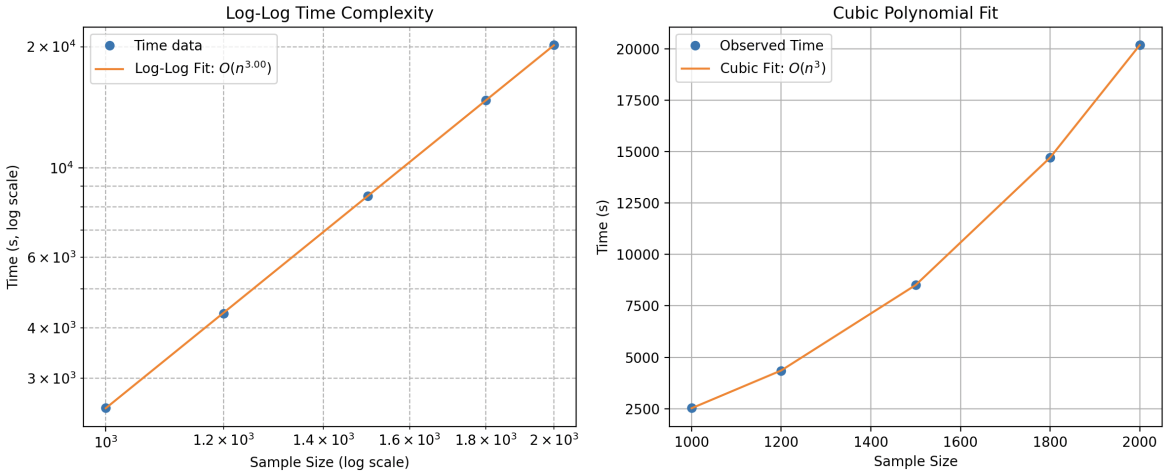


Figure 7: Codes Running Time of the First Test

The first figure shows the calculated log-log time complexity. The line in it is $\log(s) = k \cdot \log(n) + b$, *and our fitted slope* $\widehat{k}$ *is 3.00. The curve in the second figure shows the cubic polynomial fit of the theoretical running time against the sample size.*

Using the first test as an instance, we also showcase the computation complexity of our Rank MST test in Figure 7. The first figure shows a linear relationship between log(time) and log(sample size), and the estimated slope $\widehat{k}$ indicates that the empirical time complexity is $O(n^3)$, perfectly fits the theoretical time complexity. In the second figure, the observed codes running time aligns closely with the theoretical curve of $O(n^3)$, further validating its cubic dependency on sample size.

Plotting the minimal spanning tree serves as an effective way of revealing the fundamental nature of our Rank MST test.

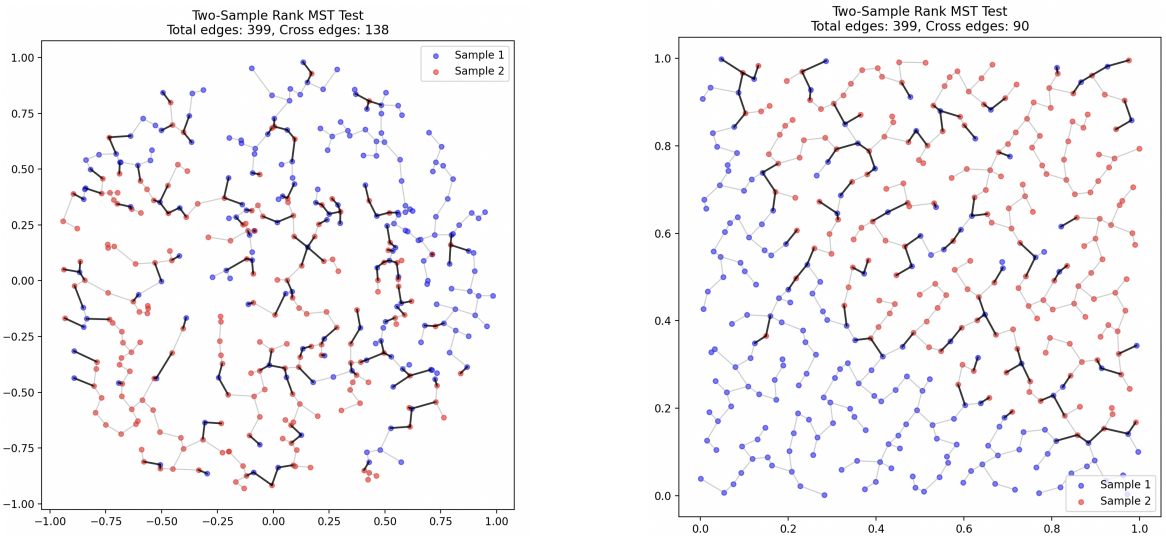


Figure 8: Minimal Spanning Trees in the Rank MST Test (Independent)

The first figure is the minimal spanning tree of the test Norm(0, 0) & Norm(-2, -2), the second figure is the minimal spanning tree of the test Norm(0, 0) & Unif(0, 6). The sample size when plotting the two figures are $m = n = 200$.

As we can see from each of the two figures, after the rank transformation, the two samples are well-separated from each other, and the number of cross sample edges, which are denoted by bolded black lines, is much smaller than half of the total sample size. This indicates clear separation between the two samples.

5.1.2 Simulation When Components Are Correlated

Now we assume the two components in each r.v. X to be correlated to each other. We generate samples from the following distributions

(b1) Norm(0, 0)$^+$: X $\sim$ N($\boldsymbol{\mu}_{b1}$, $\boldsymbol{\Sigma}_{b1}$) where $\boldsymbol{\mu}_{b1} = [0, 0]^{\mathrm{T}}$ and $\boldsymbol{\Sigma}_{b1} = \left[\begin{smallmatrix} 3 & 2 \\ 2 & 3 \end{smallmatrix}\right]$.

(b2) Norm(-1, -2)$^+$: X $\sim$ N($\boldsymbol{\mu}_{b2}$, $\boldsymbol{\Sigma}_{b1}$) where $\boldsymbol{\mu}_{b2} = [-1, -2]^{\mathrm{T}}$.

(a3) Unif(0, 6): X $\sim$ Unif$[0, 6]^2$.

(b3) T(0, 2)$^+$: X $\sim$ T($\boldsymbol{\mu}_{b3}$, $\boldsymbol{\Sigma}_{b3}$, ν) where $\boldsymbol{\mu}_{b3} = [0, 2]^{\mathrm{T}}$, $\boldsymbol{\Sigma}_{b3} = \begin{bmatrix} 1 & \frac{2}{3} \\ \frac{2}{3} & 1 \end{bmatrix}$ and $\nu = 3$.

(b4) T(1, 1)$^+$: X $\sim$ T($\boldsymbol{\mu}_{b4}$, $\boldsymbol{\Sigma}_{b3}$, ν) where $\boldsymbol{\mu}_{b4} = [1, 1]^{\mathrm{T}}$.

It can be seen that the covariance matrices above are the same.

(b5) Norm(0^+, 0^+): X $\sim$ N($\boldsymbol{\mu}_{b1}$, $\boldsymbol{\Sigma}_{b5}$, ρ) where $\boldsymbol{\Sigma}_{b5} = \begin{bmatrix} 3 & \rho \\ \rho & 2 \end{bmatrix}$ and $\rho = 1.9$.

(b6) Norm(0^+, 0^+)$^+$: X is identical to Norm(0^+, 0^+) but with the second component being randomly shuffled.

The testing results are listed as follows

GOF Testing/Methods	**Rank MST**	**Energy Distance**
Norm(0, 0)$^+$ & Norm(0, 0)$^+$	0.5342	0.5523
Norm(0, 0)$^+$ & Norm(-1, -2)$^+$	0.0010	0.0010
Norm(-1, -2)$^+$ & Unif(0, 6)	0.0010	0.0010
Norm(0, 0)$^+$ & T(0, 2)$^+$	0.0010	0.0010
T(0, 2)$^+$ & T(1, 1)$^+$	0.0010	0.0010
Unif(0, 6) & T(1, 1)$^+$	0.0010	0.0010
Norm(0^+, 0^+) & Norm(0^+, 0^+)$^+$	0.0274	0.0216

Table 2: P Values When Components Are Correlated ($m = n = 200$)

Both methods performe well in all the simulations.

Upon a detailed observation of the last test, we observe that in this setting when $\rho = 1.9$ with column shuffling, both the Rank MST and the Energy Distance test yield very small p values. This is as expected because although the shuffling of the second component in the second sample preserves marginal distributions, it severely disrupts the strong dependence structure, resulting in a substantial change in the joint distribution. Since Energy Distance is computed from pairwise Euclidean distances, such a change in the geometry of the point cloud is easily detectable by both methods.

However, the strength of our Rank MST test should become more apparent in scenarios where the change in dependence is more subtle, since its rank-based minimum spanning tree structure for capturing local and global changes in the neighborhood relationships among data points. And unlike the Energy Distance, which relies merely on Euclidean distances and is less sensitive when such distances change slightly, the Rank MST test is able to detect alterations in the underlying dependency pattern through shifts in the connectivity of the MST, even when marginal distributions and pairwise distances remain largely unchanged under the null.

To verify this, we conduct two additional simulations. The first involves changing the covariance between the two components, while the second adjusts the proportion of shuffled data in the second sample under a covariance of 1.9. By plotting the power curves under the significance level of 0.05, we can observe the respective test powers of the two methods under varying degrees of sample dependence.

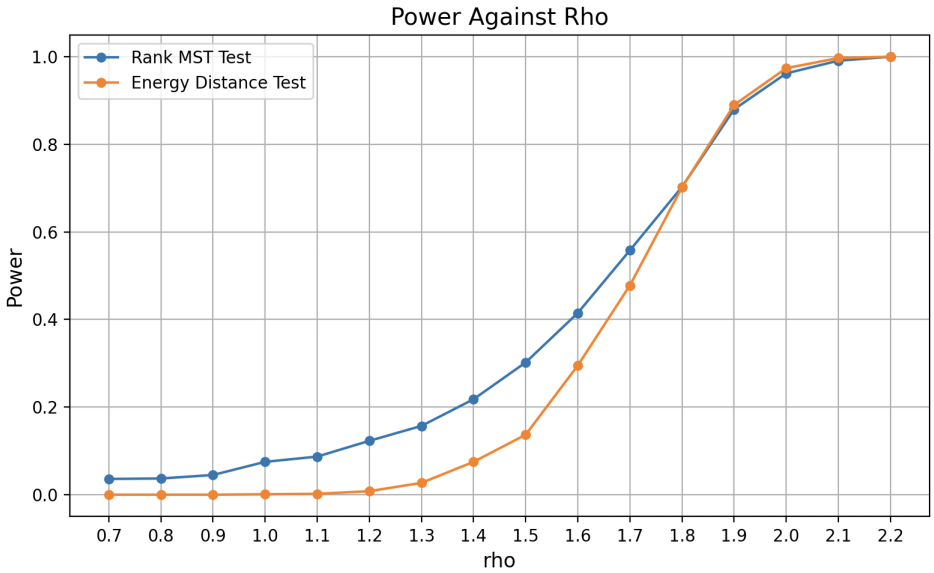


Figure 9: Power Graph of Both Methods Against ρ

This figure illustrate the power graph of both methods using the test Norm(0^+, 0^+) & Norm(0^+, 0^+)$^+$ against different ρ. Sample size of the test is still $m = n = 200$. Powers are obtained after 1000 repeated simulations.

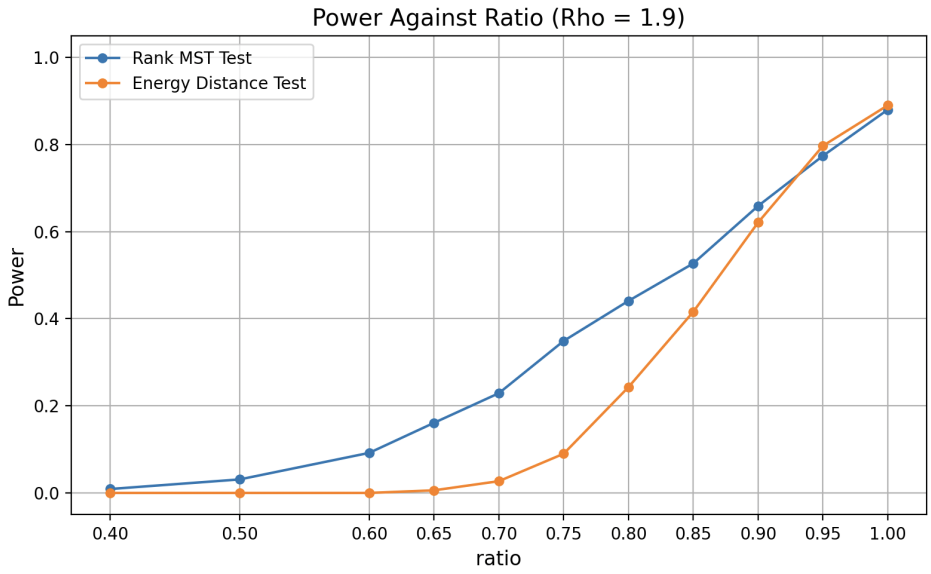


Figure 10: Power Graph of Both Methods Against Shuffling Ratio

This figure illustrate the power graph of both methods using the test Norm(0^+, 0^+) & Norm(0^+, 0^+)$^+$ against different shuffling ratios when $\rho = 0.9$. Sample size of the test is still $m = n = 200$. Powers are obtained after 1000 repeated simulations.

As shown above, when ρ (or shuffling ratio) is large, the two components are strongly correlated, and the data points are concentrated near a narrow elliptical shape or line. Shuffling the second dimension in the second sample breaks this strong dependence, causing a significant change in the joint distribution. Energy Distance, which relies on Euclidean distances, is particularly sensitive to such large geometric changes, so both the Energy test and the OT Rank MST test show high power in these cases. However, when ρ (or shuffling ratio) is small, the dependence between components weakens, and the joint distribution becomes closer to independence. Under these conditions, the geometric difference after shuffling is less obvious, making the Energy Distance test less sensitive and resulting in lower power. In contrast, the Rank MST test utilizes rank-based graph structures that are more sensitive to subtle or local changes in dependence, thus maintaining better power when ρ (or shuffling ratio) is small.

We also plot minimal spanning trees in two of our tests.

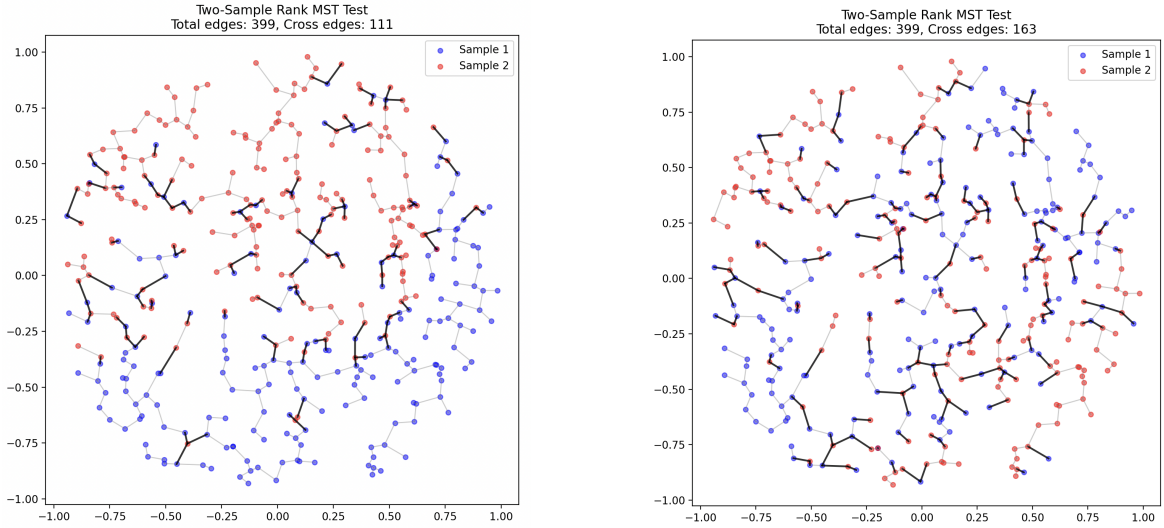


Figure 11: Minimal Spanning Trees in the Rank MST Test (Correlated)

The first figure is the minimal spanning tree of the test Norm(0, 0)$^+$ & T(0, 2)$^+$, the second figure is the minimal spanning tree of the test Norm(0$^+$, 0$^+$) & Norm(0$^+$, 0$^+$)$^+$. The sample size when plotting the two figures are $m = n = 200$.

5.1.3 Simulation of Mixed Samples

To illustrate the wide applicability of our proposed Rank MST test, we also consider simulations in the presence of sample contamination and outliers. We generate samples from the following distributions

(a1) Norm(0, 0): $\mathrm{X} \sim \mathrm{N}(\boldsymbol{\mu}_{a1}, 3\boldsymbol{I}_2)$ where $\boldsymbol{\mu}_{a1} = [0, 0]^{\mathrm{T}}$.

(c1) Norm(-12, -12)$^{++}$: $\mathrm{X} \sim \mathrm{N}(\boldsymbol{\mu}_{c2}, 3\boldsymbol{I}_2)$ where $\boldsymbol{\mu}_{c1} = [-12, -12]^{\mathrm{T}}$.

(c2) Unif(9, 15)$^{++}$: $\mathrm{X} \sim \mathrm{Unif}[9, 15]^2$.

(c3) T(12, 12)$^{++}$: X $\sim$ T($\boldsymbol{\mu}_{c3}$, $\boldsymbol{I}_2$, ν) where $\boldsymbol{\mu}_{c3} = [12, 12]^{\mathrm{T}}$ and $\nu = 3$.

(c4) Norm(24, 24)$^{++}$: X $\sim$ N($\boldsymbol{\mu}_{c4}$, $3\boldsymbol{I}_2$) where $\boldsymbol{\mu}_{c4} = [24, 24]^{\mathrm{T}}$.

In the first two tests, we focus on data contamination. We first generate a sample from a bivariate normal distribution. Then we form a second sample, with 80% of the data in it generated from the same normal distribution as above, and 20% of the data generated from a uniform distribution (and a t distribution). We aim to test whether the two methods are capable of telling the two samples apart.

In the third test, we focus on outliers in samples. We first generate a sample from a bivariate normal distribution. Then we form a second sample, with 96% of the data in it generated from the same normal distribution as above, and 4% of the data generated from a different normal distribution as outliers. We aim to test the robustness of the two methods, i.e. if they would incorrectly reject the null hypothesis.

The results are shown below and we plot minimal spanning trees in two of our tests.

GOF Testing/Methods	**Rank MST**	**Energy Distance**
Norm(0, 0) & 80% Norm(0, 0) + 20% Unif(9, 15)$^{++}$	0.0300	0.0010
Norm(0, 0) & 80% Norm(0, 0) + 20% T(12, 12)$^{++}$	0.0400	0.0010
Norm(0, 0) & 96% Norm(0, 0) + 4% Norm(24, 24)$^{++}$	0.2707	0.0010

Table 3: P Values of Mixed Samples ($m = n = 1,000$)

We can see that the smaller portion of uniform data in Sample 2 gathers at the top-right corner of the first figure. Both methods successfully distinguish between the two samples in the first two tests. In the third test, where one sample contains 4% outliers, our Rank MST test fails to reject the null hypothesis, correctly concluding that the two samples are drawn from the same distribution. In contrast, the energy distance test yields a relatively small p value, rejects the null hypothesis, and incorrectly indicates that the two samples are drawn from different distributions.

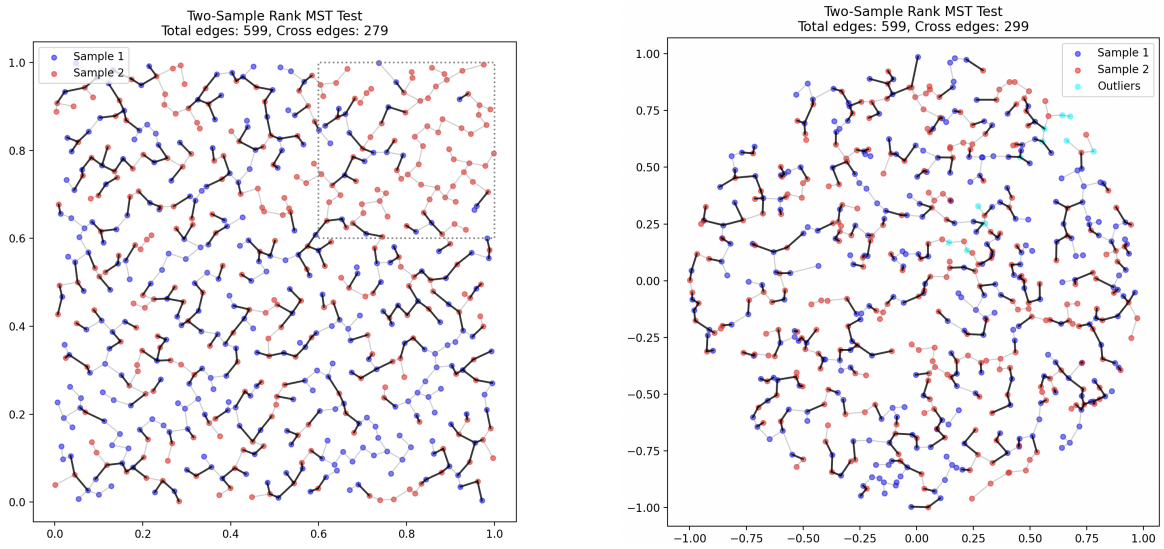


Figure 12: Minimal Spanning Trees in the Rank MST Test (Correlated)

The first figure is the minimal spanning tree of the test Norm(0, 0) & 80% Norm(0, 0) + 20% Unif(9, 15)$^{++}$, the second figure is the minimal spanning tree of the test Norm(0, 0) & 96% Norm(0, 0) + 4% Norm(24, 24)$^{++}$, with the outliers colored light blue in the topright. The sample size when plotting the two figures are $m = n = 300$.

5.2 Testing in Intermediate Dimension

By repeating the above process but with intermediate-dimensional samples, we illustrate the performance of the Rank MST test in more general situations. We aim to test under four different dimensions. We generate samples from the following distributions

(h1) Norm(0, 0): $\mathrm{X} \sim \mathrm{N}(\boldsymbol{\mu}_{h1}, 3\boldsymbol{I}_d)$ where $\boldsymbol{\mu}_{h1} = \mathbf{0}_d$.

(h2) Norm(3, 0): $\mathrm{X} \sim \mathrm{N}(\boldsymbol{\mu}_{h2}, 3\boldsymbol{I}_d)$ where $\boldsymbol{\mu}_{h2} = [3, \mathbf{0}_{d-1}^{\mathrm{T}}]^{\mathrm{T}}$.

(h3) Norm-HS(0, 0): $\mathrm{X} \sim \mathrm{N}(\boldsymbol{\mu}_{h1}, \boldsymbol{\Sigma}_{h3})$ where $\boldsymbol{\Sigma}_{h3}$ is a toeplitz matrix, and

$$\boldsymbol{\Sigma}_{h3} = 3 \times \begin{bmatrix} t_0 & t_1 & \cdots & t_{d-1} \\ t_1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & t_1 \\ t_{d-1} & \cdots & t_1 & t_0 \end{bmatrix}_{d \times d}, \quad t_i = 0.95^i$$

(h4) Unif(-3, 3): $\mathrm{X} \sim \mathrm{Unif}[-3, 3]^d$.

(h5) T(0, 0): $\mathrm{X} \sim \mathrm{T}(\boldsymbol{\mu}_{h1}, \boldsymbol{I}_d, \nu)$ where $\nu = 3$.

(h6) T(3, 0): $\mathrm{X} \sim \mathrm{T}(\boldsymbol{\mu}_{h2}, \boldsymbol{I}_d, \nu)$ where $\nu = 3$.

(h7) T(12, 0): $\mathrm{X} \sim \mathrm{T}(\boldsymbol{\mu}_{h7}, \boldsymbol{I}_d, \nu)$ where $\boldsymbol{\mu}_{h7} = [\mathbf{12}_{0.2d}^{\mathrm{T}}, \mathbf{0}_{0.8d}^{\mathrm{T}}]^{\mathrm{T}}$, $\nu = 3$.

(h8) Unif(9, 15): $\mathrm{X} \sim \mathrm{Unif}[9, 15]^{0.2d} \times [-3, 3]^{0.8d}$.

(h9) Norm(12, 0): $\mathrm{X} \sim \mathrm{N}(\boldsymbol{\mu}_{h7}, 3\boldsymbol{I}_d)$.

The testing result is shown below in the table.

GOF Testing/Dimensions	$d = 10$	$d = 20$	$d = 30$	$d = 50$
Testing of Common Samples				
Norm(0, 0) & Norm(0, 0)	0.4446	0.6653	0.7243	0.4216
Norm(0, 0) & Norm(3, 0)	0.0010	0.0010	0.0010	0.0110
Norm(3, 0) & Unif(-3, 3)	0.0010	0.0010	0.0010	0.0010
Norm(0, 0) & Norm-HS(0, 0)	0.0010	0.0010	0.0010	0.0020
Norm-HS(0, 0) & T(0, 0)	0.0010	0.0010	0.0010	0.0030
Norm(3, 0) & T(0, 0)	0.0010	0.0010	0.0030	0.0040
T(0, 0) & T(3, 0)	0.0010	0.0010	0.0010	0.0010
Testing of Mixed Samples				
Norm(0, 0) & 70% Norm(0, 0) + 30% T(12, 0)	0.0010	0.0040	0.0290	0.0679
Norm(0, 0) & 70% Norm(0, 0) + 30% Unif(9, 15)	0.0010	0.0020	0.0370	0.2288
Norm(0, 0) & 96% Norm(0, 0) + 4% Norm(12, 0)	0.8462	0.8112	0.7972	0.6653

Table 4: P Values under Different Distributions and Dimensions ($m = n = 1,000$)

In the tests when components in the r.v. are independent or correlated, our Rank MST test showcases wonderful performance in all dimensions. As for the testing of mixed samples, in cases of sample contamination, as dimension increases the power of the test decreases, but generally our test performs well, except in $\mathbb{R}^{50}$. This can be possibly explained by the increased complexity of the data space and potential "curse of dimensionality" effects that make subtle differences harder to detect.

6 Empirical Study

In this section, we apply our Rank MST test to real-world data to demonstrate how it performs in practical scenarios.

6.1 A Toy Example

The first example is the Wine Quality dataset from Cortez et al. (2009). It is a popular dataset used for regression and classification tasks. It contains information about red and white variants of Portuguese Vinho Verde wine. Each wine sample includes 11 features including fixed acidity, volatile acidity, citric acid, etc. The target variable is quality, a score between 0 and 10 given by wine tasters. The goal is usually to predict the quality score based on the chemical properties. Instead of estimating the quality variable, we conduct a goodness-of-fit test based on the other 11 physicochemical features to examine whether the two types of wine can be considered the same kind.

We randomly choose 1000 sample from each dataset, and carry out our Rank MST test. Since we are not certain whether the samples are normal-like, we try both $\text{Ball}[0,1]^{11}$ and $\text{Unif}[0,1]^{11}$ as the reference distribution.

To visualize our test, we first randomly choose 4 out of the 11 features to draw the minimal spanning tree using both reference distributions.

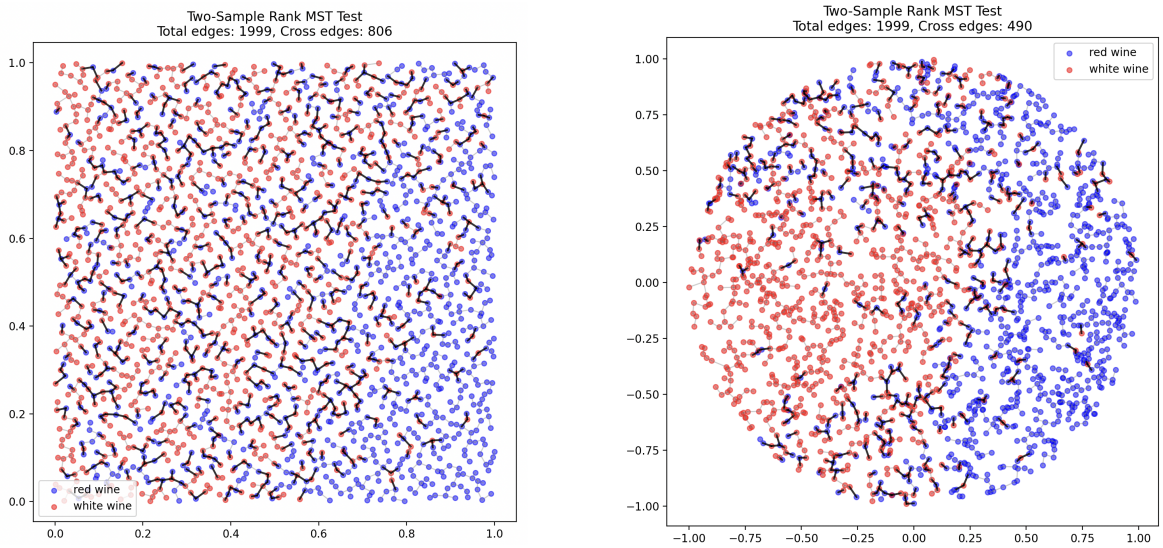


Figure 13: Minimal Spanning Trees Using the Wine Quality Dataset

The first figure is the minimal spanning tree under the reference $\text{Unif}[0,1]^2$ *using the features "free sulfur dioxide" and "total sulfur dioxide". The second figure is the minimal spanning tree under the reference* $\text{Ball}[0,1]^2$ *using the features "volatile acidity" and "citric acid".*

From the two figures, we can observe that the red and white wine datasets are clearly separated.

The result of the test is shown below

P values/References	**Unif**$[0,1]^{11}$	**Ball**$[0,1]^{11}$
P value	0.0010	0.0010

Table 5: P Values of the Wine Quality dataset ($d = 11$, $m = n = 1,000$)

Our test using both reference distribution firmly rejects the null hypothesis and correctly distinguishs two types of wine.

6.2 The Breast Cancer Dataset

The second example is the breast cancer dataset from Wasiq (2025). It contains diagnostic features for breast cancer classification, originating from the Wisconsin Breast Cancer Diagnostic study. It includes 569 samples with 30 numerical features computed from digitized images of fine needle aspirates (FNA). Each sample is labeled as either malignant (M) or benign (B). Key features in the dataset include measurements of cell nuclei characteristics like radius, texture, perimeter, and concavity, displayed as mean values, standard errors, and worst-case measurements. The dataset is clean, with no missing values, and balanced, with 357 benign and 212 malignant cases. This dataset is ideal for binary classification tasks, and is commonly used to develop machine learning models for cancer detection.

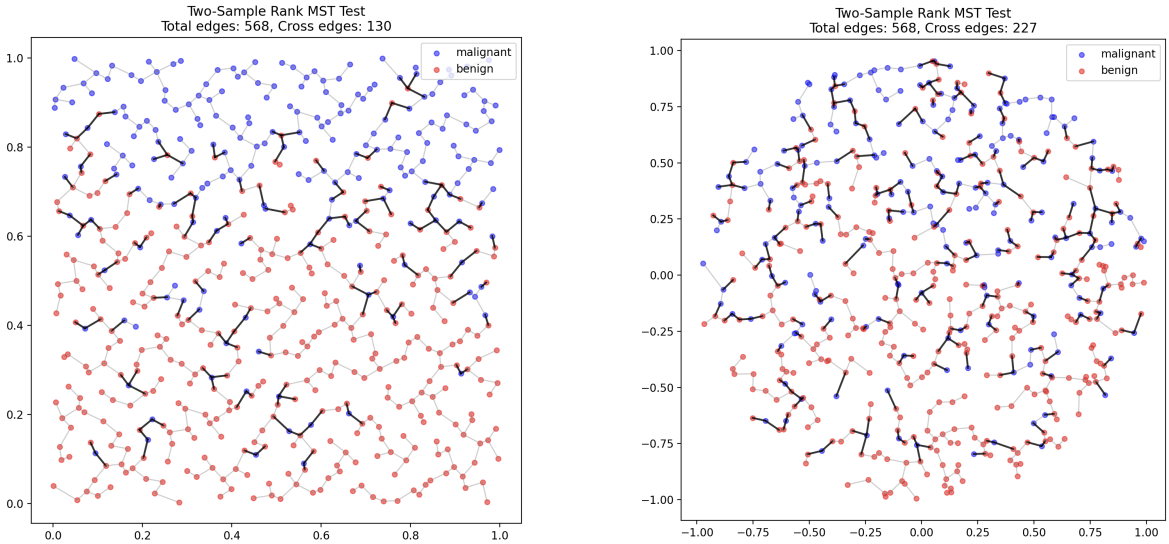


Figure 14: Minimal Spanning Trees Using the Breast Cancer Dataset

The first figure is the minimal spanning tree under the reference Unif$[0,1]^2$ *using the features "radius_mean" and "radius_se". The second figure is the minimal spanning tree under the reference Ball*$[0,1]^2$ *using the features "area_mean" and "area_se".*

To visualize our test again, we choose 4 out of the 30 features to draw the minimal spanning tree using both reference distributions, as shown above.

The result of the test is shown below

P values/References	**Unif**$[0,1]^{30}$	**Ball**$[0,1]^{30}$
P value	0.0010	0.0010

Table 6: P Values of the Wine Quality dataset ($d = 30$, $m = 357, n = 212$)

Our test using both reference distribution firmly rejects the null hypothesis and correctly distinguishs cancer and benign tumor.

7 Conclusions and Limitations

7.1 Conclusions

In this dissertation, we first offer a comprehensive literature review and comparison of several existing definitions of multivariate ranks, including component-wise ranks, spatial ranks, and depth-based ranks. While these notions extend the univariate rank concept to the multivariate scenario in different ways, they often face practical limitations such as strong dependence on the underlying distribution or non-uniqueness of rank values. These drawbacks reduce their effectiveness in using the notion of ranks to carry out nonparametric testing tasks.

To solve these issues, we adopt a new multivariate rank framework grounded in optimal transport (OT) theory. By mapping empirical data to a known reference distribution, such as the uniform or ball distribution using the Monge–Kantorovich transport map, we construct a theoretically distribution-free rank system that also performs well in finite-sample settings. Two practical ways of implementing these OT ranks are discussed in detail: one based on cell decomposition and another using Halton sequences combined with the Hungarian algorithm. The latter provides a more efficient and scalable solution for real-world problems.

Building on this OT rank foundation, we develop three novel nonparametric tests: the Rank Minimum Spanning Tree (Rank MST) test, the Rank Anderson Darling (Rank AD) test, and a multivariate version of the Wald Wolfowitz runs test. All of these methods rely on transforming the original data into rank space, where test statistics are then computed. This strategy allows us to detect differences between distributions effectively, without relying on any strong distributional assumptions.

We validate the performance of our proposed methods through extensive simulation studies across both low and intermediate dimensional settings. The Rank MST test in particular shows strong robustness and accuracy across a variety of scenarios, including cases with outliers or distributional shifts.

Compared to a widely-used benchmark, the Energy Distance test, our method performs well at capturing nonlinear patterns, handling tail differences, and managing contaminated data, and performs better in scenarios where the dependence between correlated components of random variables is subtle or partial. This results from the fact that the rank-based minimum spanning tree structure effectively detects local and global alterations in the underlying dependency structure that may not significantly affect Euclidean distances. In addition, the computational time aligns well with the expected $O(n^3)$ complexity, confirming its practical feasibility.

Overall, the OT-based multivariate ranks and the following distribution-free tests we present not only expand the set of tools available for multivariate statistical infer-

ence, but also provide a solid foundation for future work in rank-based approaches to tasks like regression, clustering, and classification.

Furthermore, we apply our Rank MST test to real-world datasets such as the Wine Quality and Breast Cancer datasets, where the test successfully distinguishes between groups with strong statistical significance under different reference distributions. These empirical results demonstrate the practical effectiveness of our methods in complex, high-dimensional settings.

7.2 Limitations

Despite both the theoretical and empirical results showing the clear advantages of OT-based multivariate ranks in nonparametric testing, several limitations remain that are worth noting:

1. Relatively high computational complexity: Although the use of the Hungarian algorithm and Halton sequences helps to improve efficiency, the overall computational complexity still scales as $O(n^3)$. This can become a bottleneck in practice when handling large-scale datasets. In fact, there are already several packages in `Python` or `R` that can be directly used to compute optimal transport. Among them, the `POT` package in `Python` implements the Sinkhorn algorithm, whose computational complexity is $O(n^2)$, which represents a significant improvement over the classical methods.

2. Dependence on the choice of reference distribution: The construction of OT-based multivariate ranks relies on a predefined reference distribution (such as the uniform or ball distribution). Different choices can lead to different geometric structures in the rank space, which may affect the sensitivity and robustness of the resulting tests. While we follow heuristic guidelines (e.g., using the ball distribution for approximately Gaussian data), there is currently no unified or theoretically justified criterion for selecting an optimal reference distribution.

3. Empirical evaluation limited to the Rank MST test: Due to page and time constraints, our simulation study focuses exclusively on the Rank MST test. Although the Rank AD test and the multivariate runs test are conceptually feasible within the same framework, their empirical performance, stability, and scope of application remain to be explored in future work. Furthermore, for the Rank MST test, the number of repetitions used to obtain the p values in our simulations is relatively limited. Ideally, one should instead evaluate the empirical power at a given significance level.

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